Demographic Parity in regression & classiffication (unawareness)

Part I
Framework from Divol and Gaucher (2024)

1 Setting

Data $(X, S, Y) \in \mathcal{X} \times \mathcal{S} \times \mathbb{R}$; $S \in \{1, \dots, K\}$.

- Group weights $p_s := \mathbb{P}(S = s)$, feature laws $\mu_s := Law(X \mid S = s)$, overall $\mu = \sum_s p_s \, \mu_s$.
- Bayes regressor (unconstrained): $\eta(x) := \mathbb{E}[Y \mid X = x]$.
- For $f : \mathcal{X} \to \mathbb{R}$, squared risk $R(f) = \mathbb{E}[(Y - f(X))^2] = \mathbb{E}[(Y - \eta(X))^2] + \mathbb{E}[(\eta(X) - f(X))^2]$.
- Pushforward: $f_{\#}\mu$ is law of $f(X)$ when $X \sim \mu$.

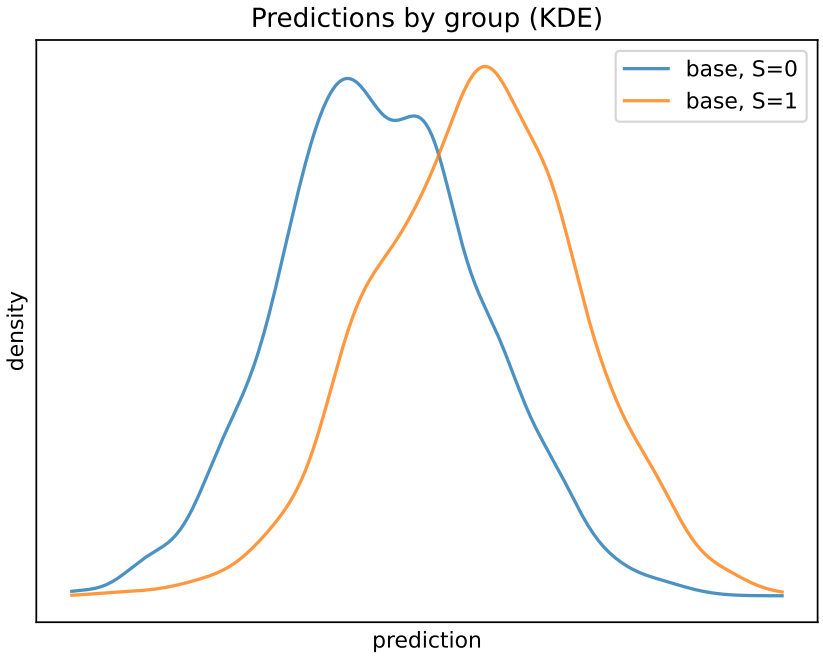


Figure 1: Distributions of $\eta(X)$ in each group (colors) and overall (black).

2 DP

DP for regression (unawareness): $f(X) \perp S$, i.e. $f_{\#}\mu_s$ is the same for all s.

$$\min_{f : \mathcal{X} \to \mathbb{R}} \mathbb{E}[(Y - f(X))^2] \quad \text{s.t.} \quad f(X) \perp S. \tag{2.1}$$

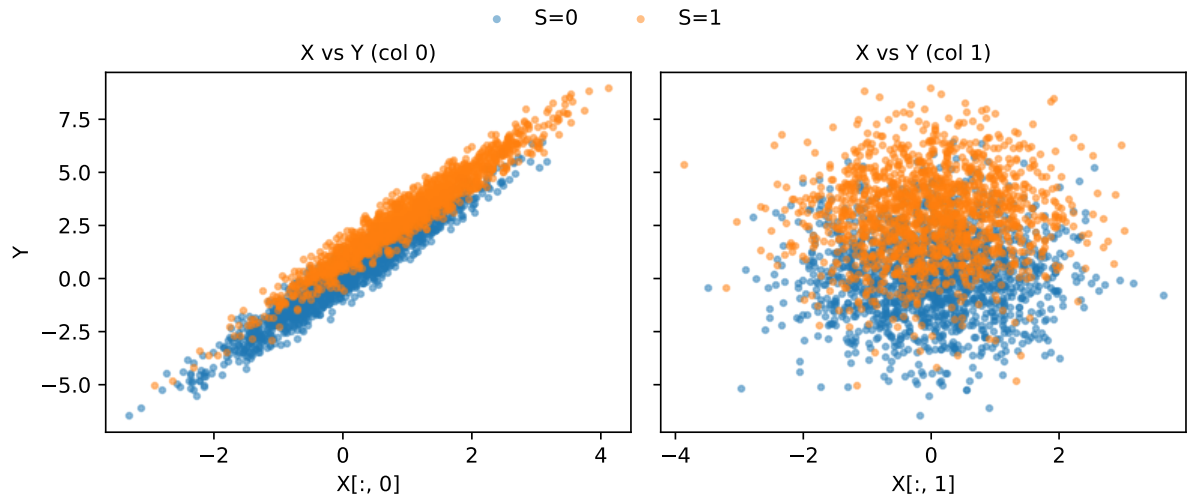


Figure 2: Scatter Y as a function of each column of X (left) on train (right) on test.

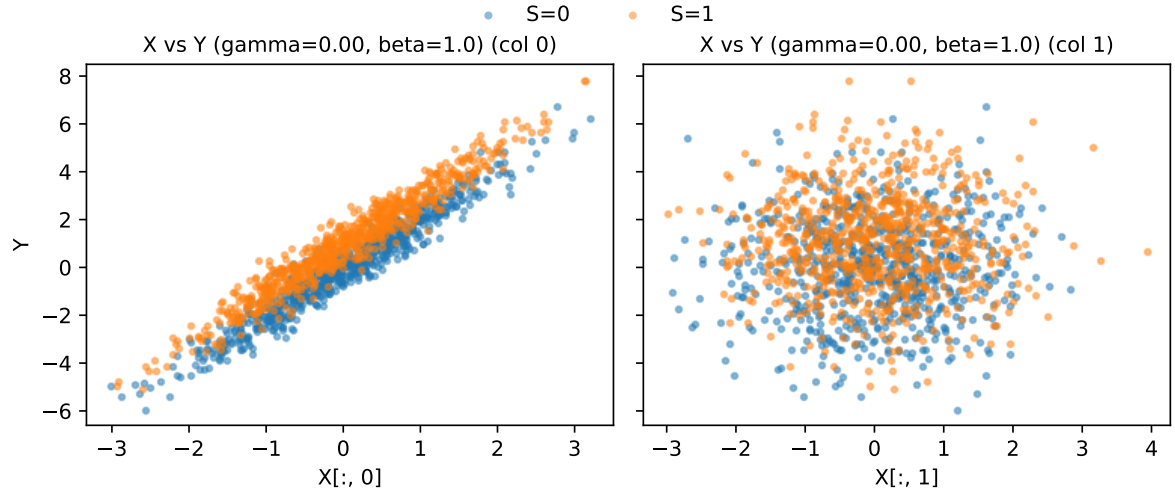


Figure 3: Scatter Y as a function of X ($\gamma = 0$) .

Order preservation (within group): if $\eta(X) < \eta(X')$ and $S = S'$, then $f(X) < f(X')$ a.s.

Key impossibility: if f has DP while η is not DP, then f cannot preserve order (proof by contrapositive: DP+↑ order ⇒ η DP).

3 Two groups ⇒ reduce to "excess-mass" pieces

Let $\alpha := \mu_1 - \mu_2$ (*signed* measure). Jordan decomposition $\alpha = \alpha^+ - \alpha^-$ with equal mass $m := \alpha^+(\mathcal{X}) = \alpha^-(\mathcal{X}) = \frac{1}{2}\|\mu_1 - \mu_2\|_{\mathrm{TV}}$.

$$\mu_+ := \alpha^+/m, \qquad \mu_- := \alpha^-/m. \tag{3.1}$$

Supports split $\mathcal{X}$ into $\mathcal{X}_+$ (where group 1 dominates), $\mathcal{X}_-$ (group 2 dominates), and the tie region $\mathcal{X}_=$.

$$\boxed{f \text{ is DP} \iff (f_{\#}\mu_+) = (f_{\#}\mu_-)} \tag{3.2}$$

(*So DP is governed by how f acts on the disagreement regions; $\mathcal{X}_=$ is irrelevant to DP.*)

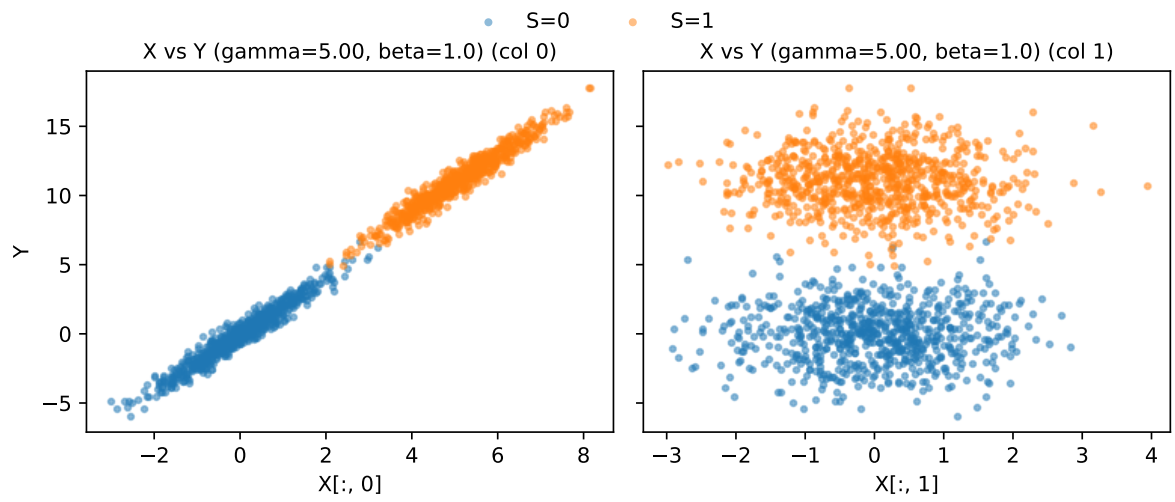


Figure 4: Scatter Y as a function of X ($\gamma = 5$) .

3. The 2D lift and the OT cost (this is the main trick)

Pick a reference μ dominating $\mu_\pm$. Define the *signed density gap*

$$\Delta(x) := \begin{cases} \dfrac{\mathrm{d}\mu_+}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_+, \\ -\dfrac{\mathrm{d}\mu_-}{\mathrm{d}\mu}(x), & x \in \mathcal{X}_-, \\ 0, & x \in \mathcal{X}_=, \end{cases} \qquad \text{and note} \quad \mathrm{d}\mu = \frac{1}{|\Delta|}\,\mathrm{d}\mu_\pm \text{ on } \mathcal{X}_\pm. \tag{3.3}$$

Lift $x \mapsto x^\uparrow := (\eta(x), \Delta(x)) = (h, d) \in \Omega := \{(h, d) \in \mathbb{R}^2 : d \neq 0\}$. Let $\mu_\pm^\uparrow$ be the laws of $x^\uparrow$ under $\mu_\pm$.

Cost (multi→one): $c((h, d), y) = \dfrac{(h - y)^2}{|d|}$.

4. Barycenter program

$$\boxed{\min_{\nu \in \mathcal{P}(\mathbb{R})} \ \mathrm{OT}_c(\mu_+^\uparrow, \nu) + \mathrm{OT}_c(\mu_-^\uparrow, \nu)} \tag{3.4}$$

This equals the optimal *excess* risk among DP regressors (first reduce risk to $\mathbb{E}[(\eta - f)^2]$, then use $\mathrm{d}\mu = \mathrm{d}\mu_\pm/|\Delta|$ on $\mathcal{X}_\pm$).

4 Collapse to a standard 2-marginal OT

For fixed (x_1, x_2) with $x_i = (h_i, d_i)$,

$$\min_y \ \frac{(h_1 - y)^2}{|d_1|} + \frac{(h_2 - y)^2}{|d_2|} \quad \Rightarrow \quad y^\star = \frac{\frac{h_1}{|d_1|} + \frac{h_2}{|d_2|}}{\frac{1}{|d_1|} + \frac{1}{|d_2|}}, \tag{4.1}$$

and minimum value

$$\boxed{C((h_1, d_1), (h_2, d_2)) = \frac{(h_2 - h_1)^2}{|d_1| + |d_2|}}. \tag{4.2}$$

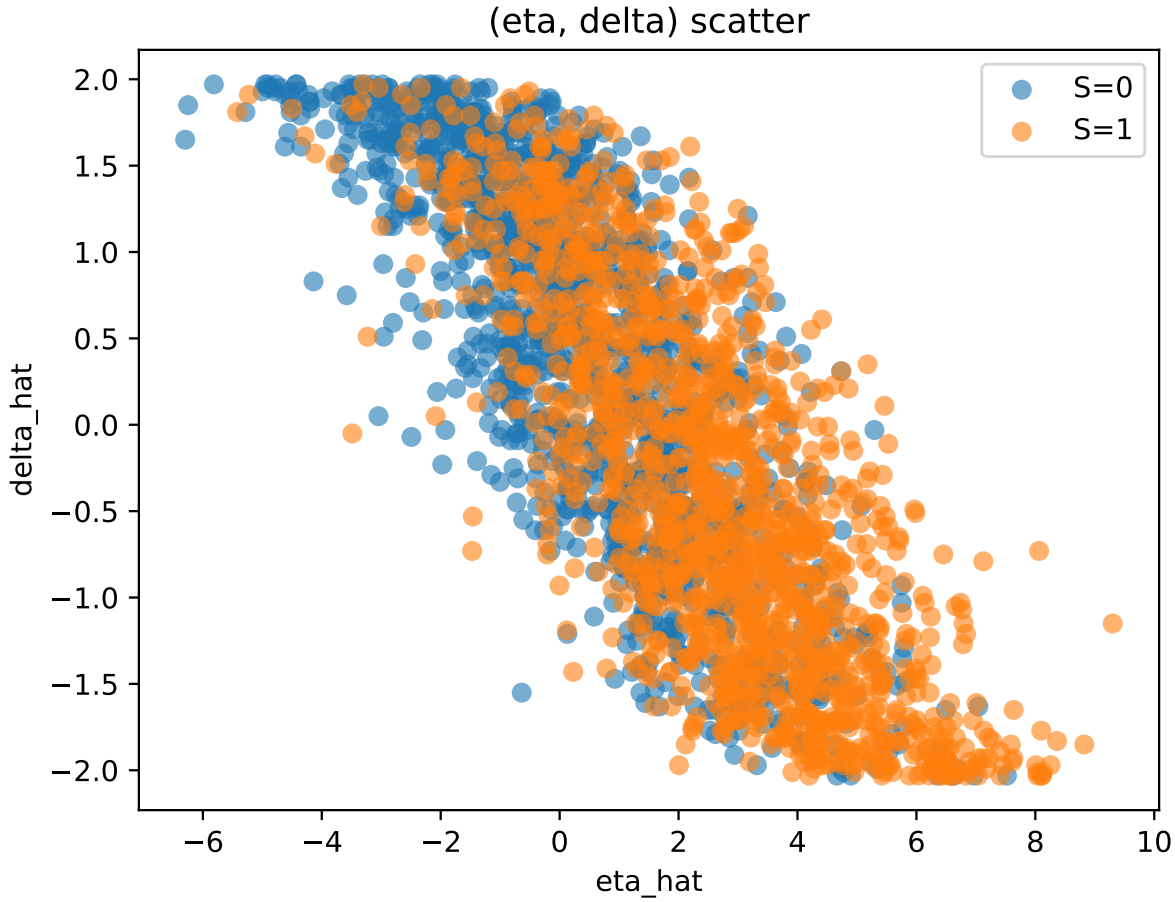


Figure 5: Scatter of (η, Δ) on train (left) and test (right).

Hence

$$\min_{\nu} \ \mathrm{OT}_c(\mu_+^{\uparrow}, \nu) + \mathrm{OT}_c(\mu_-^{\uparrow}, \nu) \ = \ \mathrm{OT}_C(\mu_+^{\uparrow}, \mu_-^{\uparrow}). \tag{4.3}$$

5 Existence, uniqueness, and maps (the mild regularity)

Assumption (zero mass on graphs). $\mu_{\pm}^{\uparrow}$ give zero measure to any set $\{(h, d) : h = F(d),\ d \neq 0\}$. (this reduces to non-atomicity assumption in the aware setting)

Under this and $\mathbb{E}[Y^2] < \infty$:

- The barycenter is *unique*: ν^{bar}.
- Each $\mathrm{OT}_c(\mu_{\pm}^{\uparrow}, \nu^{\mathrm{bar}})$ is realized by a *transport map* $f_{\pm} : \Omega \to \mathbb{R}$.

Let φ be a C-convex Kantorovich potential for $\mathrm{OT}_C(\mu_+^{\uparrow}, \mu_-^{\uparrow})$. Where $\partial_h \varphi(h, d)$ exists,

$$\boxed{f_+(h, d) = h - \frac{|d|}{2}\, \partial_h \varphi(h, d)} \qquad \text{(and symmetrically } f_-\text{)}. \tag{5.1}$$

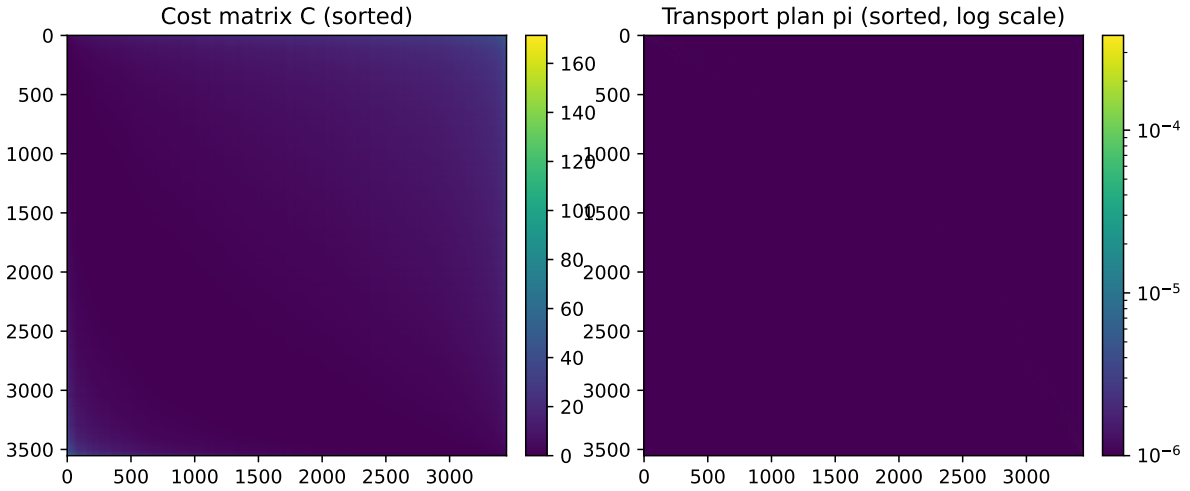


Figure 6: OT matrix between $\mu_+^{\uparrow}$ and $\mu_-^{\uparrow}$.

Subset	Spearman ρ	Kendall τ_b	Mean $\lvert\Delta\text{rank}\rvert$	Top-5% Jaccard
overall	0.662	0.509	0.216	0.250
S=0	0.980	0.883	0.044	0.682
S=1	0.979	0.881	0.045	0.805

Table 1: Ordering stability between $\hat{\eta}$ and corrected scores.

7. Final form of the optimal fair regressor

$$f^\star(x) = \begin{cases} f_+\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_+, \\ f_-\big(\eta(x), \Delta(x)\big), & x \in \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases} \qquad \text{and} \qquad (f^\star)_\# \mu_+ = (f^\star)_\# \mu_- = \nu^{\text{bar}}. \tag{5.2}$$

6 Nestedness (multi to one OT)

Setup: source $(X, \mu) \subset \mathbb{R}^m$, target $(Y, \nu) \subset \mathbb{R}$, cost $c(x, y)$ differentiable in y. Define sublevel sets using the *marginal cost*

$$X_{\le}(y, k) := \{x \in \mathcal{X} : \partial_y c(x, y) \le k\}. \tag{6.1}$$

For each y, choose $k(y)$ so that $\mu\big(X_{\le}(y, k(y))\big) = F_\nu(y)$ where F_ν is the CDF of ν.

Nestedness:

$$y_1 < y_2 \implies X_{\le}\big(y_1, k(y_1)\big) \subseteq X_{\le}\big(y_2, k(y_2)\big). \tag{6.2}$$

Consequence: existence of a Monge map $T : \mathcal{X} \to Y$ with $\partial_y c\big(x, T(x)\big) = k\big(T(x)\big)$ a.e. (no mass splitting).

Tiny example. If $c(x, y) = \frac{1}{2}\|x\|^2 - y\, x_1$, then $\partial_y c(x, y) = -x_1$, so $X_{\le}(y, k) = \{x_1 \ge -k\}$ (independent of y). Picking $k(y)$ to match $F_\nu(y)$ makes these sets nested $\Rightarrow$ a map depending on x_1 only.

$\implies$ mélange entre les groupes mais dans chaque groupe l'ordre est assez preservé

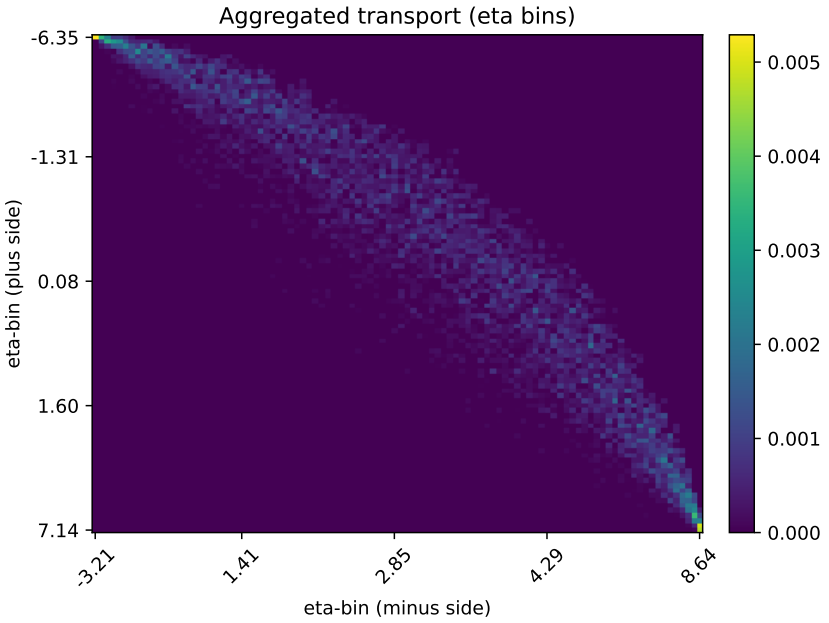
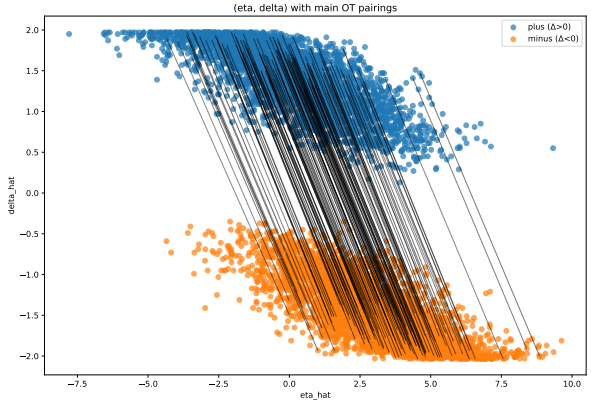
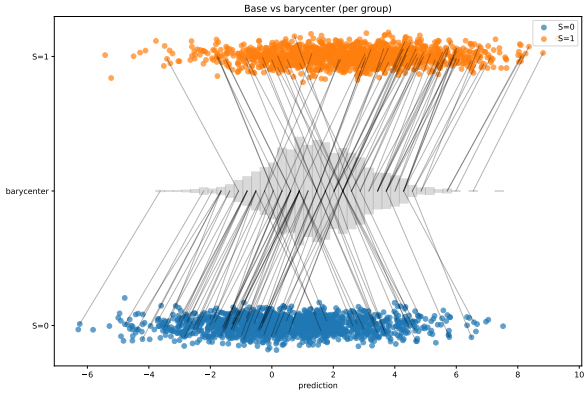


Figure 7: Aggregated OT plan projected on the η-axis.



OT transport plan from $\mu_+^{\uparrow}$ to $\mu_-^{\uparrow}$ (top 30% largest moves; max 150 links).



Barycenter links from (η, Δ) to the fair prediction (top 30% largest moves; max 150 links).

Figure 8: OT transport plan (left) VS barycenter links (right).

7 Classification link (cost-sensitive, $Y \in \{0, 1\}$)

For weight $y \in [0, 1]$,

$$R_y(g) = y\,\mathbb{P}(Y = 0,\, g(X) = 1) + (1 - y)\,\mathbb{P}(Y = 1,\, g(X) = 0), \quad g : \mathcal{X} \to \{0, 1\}. \tag{7.1}$$

Under DP and a *nestedness* condition, the optimal fair classifiers are *thresholds* of $f^\star$:

$$g_y^\star(x) = \mathbf{1}\{\, f^\star(x) \geq t(y)\,\}, \tag{7.2}$$

and the family $\{g_y^\star\}_y$ recovers $f^\star$ (up to null sets). Without nestedness this equivalence can fail.

8 Awareness VS unawareness

- **Awareness** (can use S): DP-optimal regressor is a *groupwise* OT/quantile match to a common barycenter; preserves within-group order; often closed-form via CDF↔quantile.

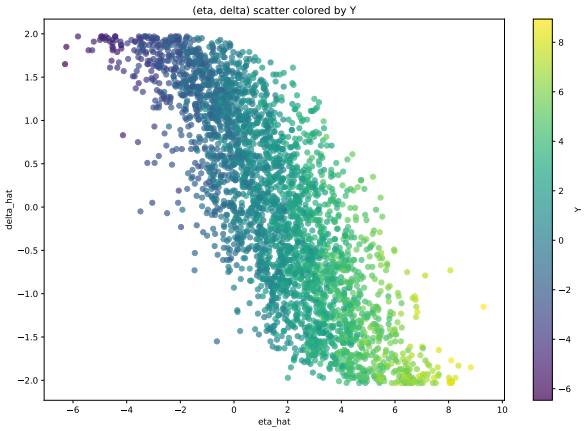


Figure 9: Train on the left test on the right

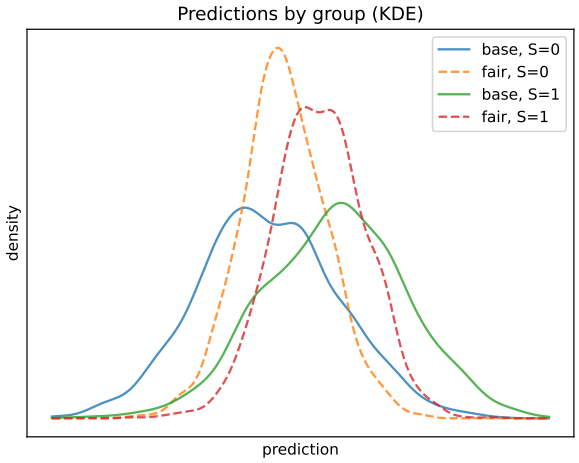


Figure 10: Distributions before and after correction in each group.

Quantity	AUC
$\mathrm{AUC}(S \mid Y)$	0.858
$\mathrm{AUC}(S \mid \hat{\eta})$	0.828
$\mathrm{AUC}(S \mid \tilde{y})$	0.396

Table 2: How easy it is to infer S from te score

- **Unawareness**: DP forces a *2D→1D* OT barycenter depending on (η, Δ); no general closed form; order preservation can *fail*.

9 In practice

1. Estimate $\widehat{\eta}(x)$ and $\widehat{\Delta}(x)$ (need $\widehat{\mathbb{P}}(S = s \mid X = x)$ or density ratios).
2. Build empirical $\widehat{\mu}_{\pm}^{\uparrow}$ on samples from $\mathcal{X}_{\pm}$.
3. Solve OT_C with $C((h_1, d_1), (h_2, d_2)) = (h_2 - h_1)^2/(|d_1| + |d_2|)$ to get maps $\hat{f}_{\pm}$.
4. Output $\hat{f}^{\star}$ piecewise on $(\mathcal{X}_+, \mathcal{X}_-, \mathcal{X}_=)$ as above.

10 How good is the method when the X by S are more or less confounded?

All confounded experiements are with $n = 1500$

11 defs

- **DP:** $f(X) \perp S \iff f_{\#}\mu_s$ independent of s.
- **Pushforward:** $f_{\#}\mu(A) = \mu(f^{-1}(A))$.
- **Jordan decomposition:** $\alpha = \alpha^- - \alpha^+$ with disjoint supports, $|\alpha| = \alpha^+ + \alpha^-$.
- **Barycenter (here):** ν minimizing OT-distance sum from sources.
- **Nestedness:** monotone growth in y of $\partial_y c$-sublevel sets with prescribed μ-mass.
- **Kantorovich potential:** dual function whose subgradient characterizes optimal plans/maps.

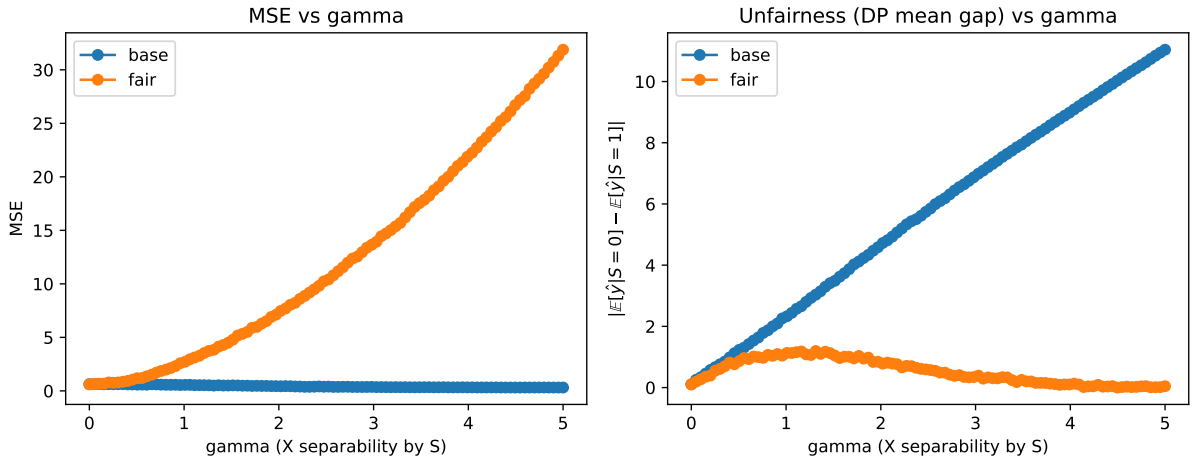


Figure 11: Results for different gamma values (make sense bc when $\gamma = 0$ fairness is free).

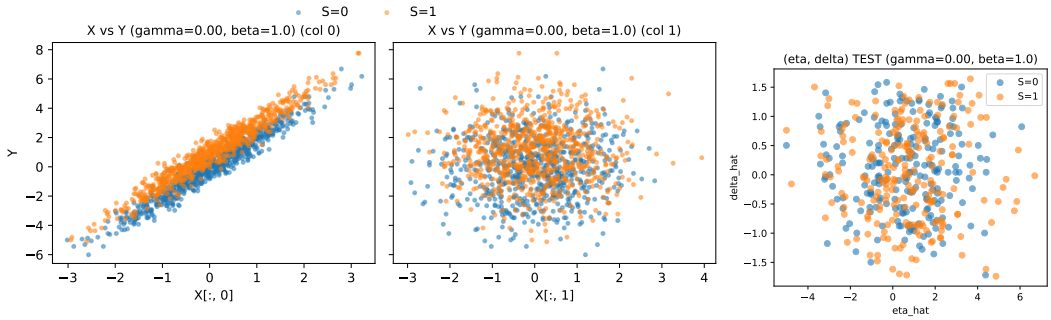


Figure 12: Results at $\gamma = 0$: scatter (X, Y) vs. (η, Δ) side by side.

12 formuless

$$\boxed{c((h,d),y) = \frac{(h-y)^2}{|d|}}, \qquad \boxed{C((h_1,d_1),(h_2,d_2)) = \frac{(h_2-h_1)^2}{|d_1|+|d_2|}}, \tag{12.1}$$

$$\boxed{f^\star(x) = \begin{cases} \eta(x) - \dfrac{|\Delta(x)|}{2}\,\partial_h\varphi(\eta(x),\Delta(x)), & x \in \mathcal{X}_+ \cup \mathcal{X}_-, \\ \eta(x), & x \in \mathcal{X}_=. \end{cases}}. \tag{12.2}$$

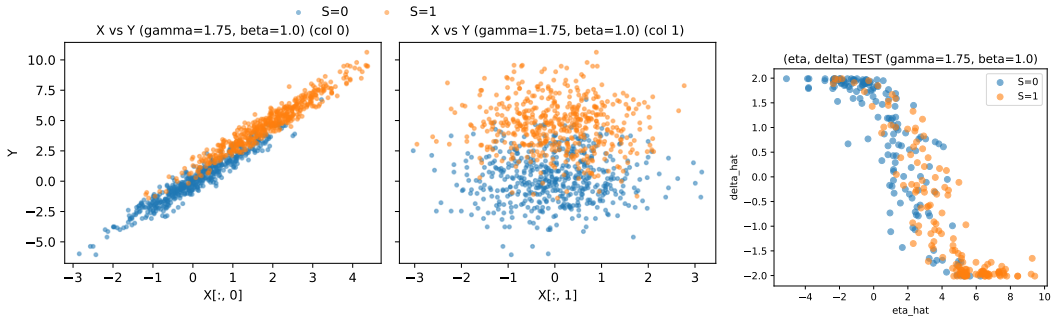


Figure 13: Results at $\gamma = 1.88$: scatter (X, Y) vs. (η, Δ) side by side.

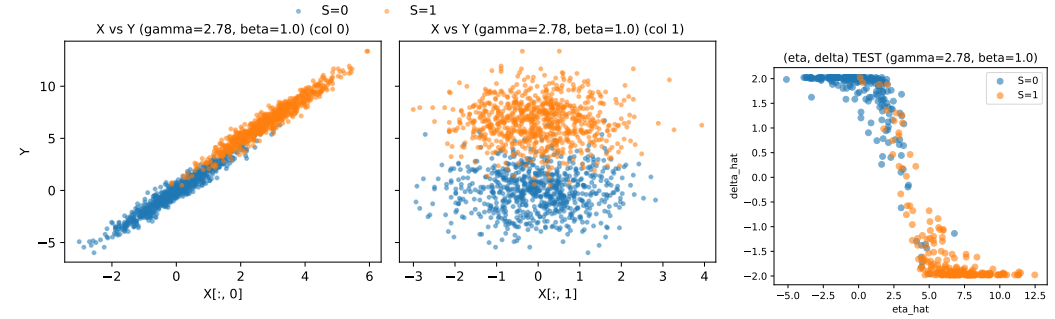


Figure 14: Results at $\gamma = 2.50$: scatter (X, Y) vs. (η, Δ) side by side.

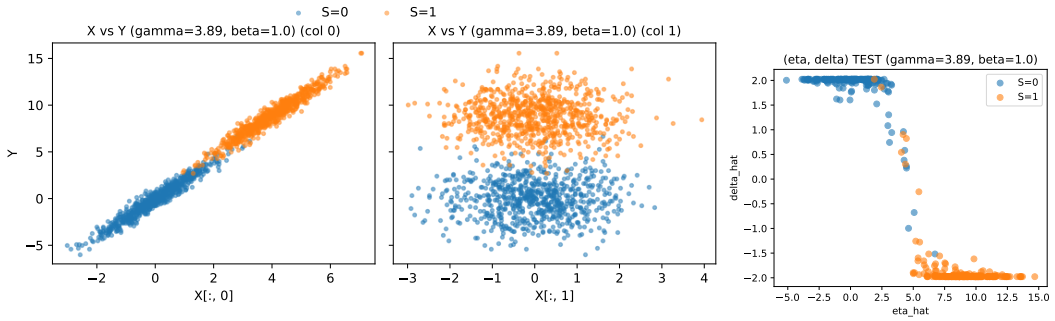


Figure 15: Results at $\gamma = 3.75$: scatter (X, Y) vs. (η, Δ) side by side.

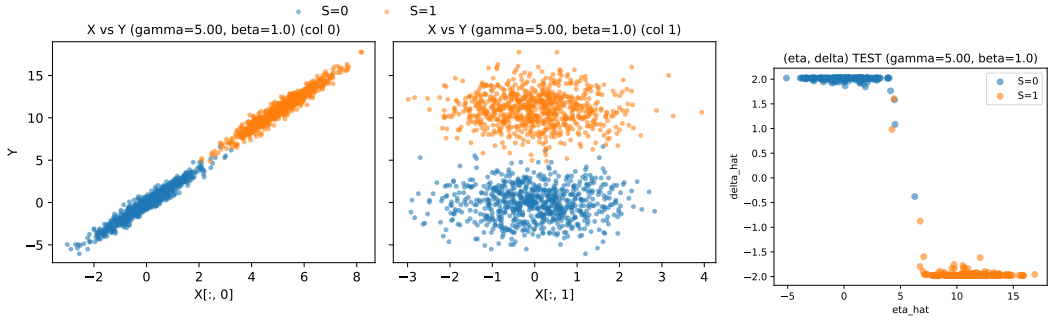


Figure 16: Results at $\gamma = 5.00$: scatter (X, Y) vs. (η, Δ) side by side.

Part II
Unawareness from Chzhen, Hebiri, and Taturyan (2024)

1 what porblem do they solve?

We want regression under **demographic parity (DP)** in the **unawareness** setting: the predictor cannot use S at inference time, but we still want $\widehat{Y} \perp S$.

Key design choice: build a *smooth convex dual objective* whose stochastic gradients can be computed from *unlabeled* X only, enabling **online post-processing** on a stream of i.i.d. X.

2 Setting and notation

Data $(X, S, Y) \in \mathbb{R}^d \times [K] \times \mathbb{R}$ with unknown law P.

- Regression function: $\eta(x) := \mathbb{E}[Y \mid X = x]$.
- Sensitive marginals: $p_s := \mathbb{P}(S = s)$, and vector $p = (p_s)_{s\in[K]}$.
- Sensitive predictor / propensity: $\tau(x) := (\tau_s(x))_{s\in[K]}$ with $\tau_s(x) := \mathbb{P}(S = s \mid X = x)$.

Randomized predictors (unawareness). A randomized prediction function is a Markov kernel

$$\pi : \begin{cases} \mathcal{B}(\mathbb{R}) \times \mathbb{R}^d & \to [0,1], \\ B & \mapsto \pi(B \mid x). \end{cases} \tag{2.1}$$

It induces a prediction random variable $\widehat{Y}_\pi$ by

$$Law(\widehat{Y}_\pi \mid X = x, S = s) = \pi(\cdot \mid x). \tag{2.2}$$

Remark 2.1: deterministic predictors correspond to Dirac kernels $\pi(\cdot \mid x) = \delta_{g(x)}$, giving $\widehat{Y}_\pi = g(X)$. Also, π specifies the joint law of $(X, S, \widehat{Y}_\pi)$, and to relate it to (X, S, Y) one may assume conditional independence $\widehat{Y}_\pi \perp Y \mid (X, S)$ so that $(X, S, Y, \widehat{Y}_\pi)$ is well-defined.

3 Demographic parity and risk

DP. A predictor π satisfies demographic parity if

$$\widehat{Y}_\pi \perp S. \tag{3.1}$$

This is *distributional*, not pointwise in x.

Risk. They measure performance against η (excess risk):

$$R(\pi) := \mathbb{E}\big[(\widehat{Y}_\pi - \eta(X))^2\big] = \mathbb{E}\Big[\int_{\mathbb{R}} (\widehat{y} - \eta(X))^2 \, \pi(d\widehat{y} \mid X)\Big]. \tag{3.2}$$

4 discretization of the prediction range

They force the predictor outputs to lie on a **fixed finite grid**.

Bounded signal assumption. Assume there exists $B > 0$ such that $|\eta(X)| \le B$ a.s.

Grid. For an integer $L \geq 1$, define the uniform grid on $[-B, B]$

$$\widehat{\mathcal{Y}}_L := \left\{ \widehat{y}_\ell := \frac{\ell B}{L} : \ell \in [[L]] := \{-L, \dots, L\} \right\}, \qquad |\widehat{\mathcal{Y}}_L| = 2L + 1. \tag{4.1}$$

A feasible predictor must satisfy $Supp(\pi(\cdot \mid x)) = \widehat{\mathcal{Y}}_L$ for all x.

Why discretize? Because with a fixed finite support, DP reduces to finitely many *linear* constraints on the probabilities $\pi(\ell \mid x)$:

$$\mathbb{P}(\widehat{Y}_\pi = \widehat{y}_\ell \mid S = s) = \mathbb{P}(\widehat{Y}_\pi = \widehat{y}_\ell) \quad \Longleftrightarrow \quad \mathbb{E}[\pi(\ell \mid X) \mid S = s] = \mathbb{E}[\pi(\ell \mid X)]. \tag{4.2}$$

They allow slack via thresholds $\varepsilon = (\varepsilon_s)_{s \in [K]}$ and define (per group, per bin)

$$U_s(\pi, \ell) := \Big| \mathbb{E}[\pi(\ell \mid X) \mid S = s] - \mathbb{E}[\pi(\ell \mid X)] \Big| \leq \varepsilon_s. \tag{4.3}$$

Important sanity point (parce que j'ai eu un doute): this is *not* turning regression into classification. The algorithm learns a *distribution* over grid values; you can output either a sample $\widehat{Y}_\pi$ or the mean $\mathbb{E}[\widehat{Y}_\pi \mid X = x] = \sum_\ell \widehat{y}_\ell \, \pi(\ell \mid x)$, which is still a real-valued regressor. The discretization is an algorithmic device to make DP constraints finite.

5 Entropic regularization ⇒ smooth dual convex program

Solving the discretized constrained problem directly is a large LP in dimension $\Omega(LK)$, so they switch to an entropic-regularized variant (comme c'est leur gros argument je pense que dans la comparaison il faudra voir la complexité temporelle *a minima*).

Entropy-regularized risk. Let $\Psi(\mu) = \sum_{y \in Supp(\mu)} \mu(y) \log \mu(y)$ denote negative entropy. For $\beta > 0$:

$$R_\beta(\pi) := R(\pi) + \frac{1}{\beta} \mathbb{E}\big[\Psi(\pi(\cdot \mid X))\big]. \tag{5.1}$$

Dual variables and key reparameterization. Define

$$t(x) := \mathbf{1} - \frac{\tau(x)}{p} \in \mathbb{R}^K \qquad \text{(componentwise division)}, \qquad r_\ell(x) := \big(\eta(x) - \widehat{y}_\ell\big)^2. \tag{5.2}$$

5.1 Lemma 3.1 (the main structural lemma)

The entropic-regularized problem admits a solution that can be written explicitly in terms of nonnegative dual matrices $\Lambda = (\lambda_{\ell s})$ and $V = (\nu_{\ell s})$, as follows. 2 matrices parce que contrainte en valeur absolue

Dual objective (smooth convex). Let $\lambda_\ell = (\lambda_{\ell s})_{s \in [K]}$ and $\nu_\ell = (\nu_{\ell s})_{s \in [K]}$. Define the log-sum-exp at temperature β by

$$\mathrm{LSE}_\beta(w) = \beta^{-1} \log \Big(\sum_j e^{\beta w_j} \Big), \tag{5.3}$$

and the soft-argmax / softmax by $\sigma_j(w) = \frac{e^{w_j}}{\sum_i e^{w_i}}$. Then define

$$F(\Lambda, V) := \mathbb{E}\Big[\mathrm{LSE}_\beta\Big(\big(\langle \lambda_\ell - \nu_\ell, \, t(X) \rangle - r_\ell(X) \big)_{\ell \in [[L]]} \Big) \Big] \; + \sum_{\ell \in [[L]]} \langle \lambda_\ell + \nu_\ell, \varepsilon \rangle, \quad \Lambda, V \geq 0. \tag{5.4}$$

Primal solution (softmax form). If $(\Lambda^\star, V^\star)$ minimizes F, then an optimal predictor is

$$\boxed{\pi_{\Lambda^\star, V^\star}(\ell \mid x) = \sigma_\ell\Big(\beta\big(\langle \lambda^\star_{\ell'} - \nu^\star_{\ell'},\, t(x)\rangle - r_{\ell'}(x)\big)_{\ell' \in [[L]]}\Big)\,.} \tag{5.5}$$

Interpretation. When $\Lambda = V = 0$, (no fairness constraints), we have

$$\pi_{0,0}(\ell \mid x) \propto \exp\big(-\beta(\eta(x) - \widehat{y}_\ell)^2\big), \tag{5.6}$$

so the distribution concentrates around the grid point closest to $\eta(x)$; the dual terms then reweight bins to enforce DP across groups.

5.2 Regularity + why gradient norm matters (Lemma 3.4 + 3.5)

Let

$$\sigma^2 := 2 \sum_{s \in [K]} \frac{1 - p_s}{p_s}. \tag{5.7}$$

Then ∇F is $(\beta\sigma^2)$-Lipschitz, so larger β makes F less smooth.

Since the dual domain is constrained ($\Lambda, V \geq 0$), they use the **gradient mapping** (for stepsize $\alpha > 0$)

$$G_\alpha(\Lambda, V) := \frac{(\Lambda, V) - \big((\Lambda, V) - \alpha \nabla F(\Lambda, V)\big)_+}{\alpha}, \tag{5.8}$$

where $(\cdot)_+$ is coordinatewise projection onto $\mathbb{R}_+^{2(2L+1)K}$.

Lemma 3.5 (key implication). For any $(\Lambda, V) \geq 0$, the *fairness violation* of $\pi_{\Lambda,V}$ is controlled by $\|G_\alpha(\Lambda, V)\|$, and the *risk suboptimality* is bounded by a term linear in $\|G_\alpha(\Lambda, V)\|$ plus the entropic bias $\log|\widehat{\mathcal{Y}}_L|/\beta$.

Moral: they need algorithms that directly control $\mathbb{E}\|G_\alpha\|^2$ with clean statistical rates.

6 Algorithm: online unlabeled post-processing (Algorithm 1)

Inputs: L, β, number of stochastic gradient calls T, p, B, and functions η, τ.

1. Build grid $\widehat{\mathcal{Y}}_L$.
2. Define $F(\Lambda, V)$ as above; set smoothness proxy $M = \beta\sigma^2$.
3. Run a black-box stochastic optimizer $\mathcal{A}$ on F for T stochastic gradients to get $(\widehat{\Lambda}, \widehat{V})$.
4. Output $\pi_{\widehat{\Lambda}, \widehat{V}}$ via the softmax formula.

Stochastic gradient oracle uses only X. They derive ∇F explicitly and estimate it by sampling a single $X \sim P_X$ and removing the expectation. So the post-processing step can be done on a **stream** of unlabeled X without storing data.

Unknown η and τ. Train $\widehat{\eta}$ and $\widehat{\tau}$ on a labeled dataset, then run the same post-processing on an independent unlabeled stream; additional error terms appear that depend on the quality of $\widehat{\eta}, \widehat{\tau}$.

7 Main guarantees (Theorem 5.1 + plugin extension)

They choose parameters as functions of the number of unlabeled samples/iterations T (e.g. $L \asymp \sqrt{T}$ and a matching β schedule) and prove that:

- fairness violation decreases at a rate essentially $1/\sqrt{T}$ (up to logs and constants),
- excess risk (vs. a natural deterministic DP benchmark based on Kolmogorov–Smirnov-type constraints) also decreases like $1/\sqrt{T}$,
- dependence on K appears through σ and on constraint tightness through $\|(\Lambda^\star, V^\star)\|$.

They also provide post-processing-style bounds for the plug-in version with $(\widehat{\eta}, \widehat{\tau})$, separating estimation error from optimization/post-processing error.

8 what is actually run

Datasets:

- Law School: predict GPA (scaled to $[0, 1]$),
- Communities & Crime: predict violent crime rate (scaled to $[0, 1]$),

with ethnicity as sensitive attribute (binary: white vs non-white).

Pipeline: split into train / unlabeled / test (0.4 / 0.4 / 0.2); train $\widehat{\eta}$ via LinearRegression and $\widehat{\tau}$ via LogisticRegression; run Algorithm 1 using the unlabeled split; evaluate risk and unfairness vs iterations.

They compare against Agarwal–Dudik–Wu (2019)in two training regimes (with and without access to unlabeled data) $\implies$ competitive or better tradeoffs.

9 Formules

- Grid indices: $[[L]] = \{-L, \dots, L\}$, grid values $\widehat{y}_\ell = \ell B/L$.
- Slack DP constraints: $U_s(\pi, \ell) \le \varepsilon_s$.
- Dual objective: $F(\Lambda, V) = \mathbb{E}[\mathrm{LSE}_\beta(\cdots)] + \sum_\ell \langle \lambda_\ell + \nu_\ell, \varepsilon \rangle$.
- Output predictor: $\pi_{\Lambda,V}(\ell \mid x) = \mathrm{softmax}_\ell\big(\beta(\langle \lambda_{\ell'} - \nu_{\ell'}, t(x) \rangle - r_{\ell'}(x))\big)$.
- Gradient mapping: $G_\alpha(\Lambda, V) = \frac{(\Lambda,V) - ((\Lambda,V) - \alpha \nabla F)_+}{\alpha}$.

10 Comparison: OT characterization vs online convex post-processing

One-line takeaway

- **Divol–Gaucher:** population-level characterization of the optimal DP regressor via an OT barycenter on a lifted space (η, Δ) (mainly two groups).
- **Taturyan–Chzhen–Hebiri:** practical post-processing algorithm with finite-sample bounds, based on discretization + entropic smoothing + stochastic convex optimization, usable with an unlabeled stream of X.

Differences

	OT	**Unlabeled post-processing**
Primary goal	Characterize $f^\star$ solving $\min_f \mathbb{E}[(Y - f(X))^2]$ s.t. $f(X) \perp S$ (population optimum).	IMPLEMENTABLE estimator achieving (approx.) DP with rates, using post-processing.
Groups	Focuses on $S \in \{1, 2\}$ (extension to $K > 2$ left for future work).	Handles $S \in [K]$ and discusses multiple protected attributes.
Form of predictor	Deterministic regressor $f^\star : \mathcal{X} \to \mathbb{R}$, described via OT on a lift $(\eta(x), \Delta(x))$.	Randomized predictor $\pi(\cdot \mid x)$ supported on a **grid** $\widehat{\mathcal{Y}}_L$ (can output a sample or the mean).
How DP is enforced	**Exact** DP constraint at the population level ($f(X) \perp S$).	**Relaxed / discretized** DP constraints on grid atoms (with slack ε); plus entropic regularization.
Key mathematical object	OT barycenter with a *multi-to-one* cost after lifting; DP governed by "disagreement" regions and a signed density gap Δ.	Smooth convex dual objective $F(\Lambda, V)$; optimal π is an explicit **softmax** (Lemma 3.1).
What information is implicitly needed	Needs $\eta(x)$ and $\Delta(x)$, where Δ depends on group-membership probabilities $\mathbb{P}(S = s \mid X = x)$ (implicit use of S).	Needs $\eta(x)$ and $\tau(x) = (\mathbb{P}(S = s \mid X = x))_s$ to form $t(x) = \mathbf{1} - \tau(x)/p$.
Computation / "online"	typically offline	Stochastic convex optimization with an oracle computable from *unlabeled* i.i.d. samples of X; suited for streaming.

Fainress through $\Delta(x)$ or $\tau(x)$.

- **OT characterization** more interpretable: what the optimal $f^\star$ looks like, why order-preservation can fail, and how DP reshapes predictions geometrically.
- **unlabeled post-processing** more applicable *pipeline*: a concrete algorithm with controllable approximation (L, β, ε) and finite-sample/optimization guarantees.

$\implies$ interesting to see the statiqsticak guarantees we can have with the OT post-processing + see what happens with $K > 2$.

Part III
Comparison on real data

J'ai importé leur code pour utiliser exactement la meme chose, j'ai testé plein de truc mais j'ai toujours des matmul warning de l'enfer meme en scalant tout par contre j'ai bien verifié et toutes les données sont finies donc c'est harmless en terme de perf mais c'est bizarre

Data splitting and sample sizes. After loading (X, S, y), we follow the same preprocessing choices as the original FairReg evaluation code. In particular, for `adult` and `lawschool` we optionally rescale the target y to $[-1, 1]$ (via a min–max transform) and we subsample the dataset to a fixed size of $N = 2000$ points. For `communities`, we keep the full dataset (denote by N its size after loading and optional scaling). For each run i, we then construct three disjoint splits using two successive calls to `train_test_split`, always stratified by the sensitive attribute S to preserve group proportions. First, we split the full dataset into a training set of size $1 - \texttt{TEST_SIZE}$ and a remaining set of size `TEST_SIZE`. Second, we split this remaining set into an unlabeled set and a test set, using

$$\texttt{train_size} = \frac{\texttt{TRAIN_SIZE}}{1 - \texttt{TEST_SIZE}}.$$

With the default parameters $\texttt{TRAIN_SIZE} = 0.4$ and $\texttt{TEST_SIZE} = 0.2$, this yields the following proportions of the total sample size N:

$$\begin{aligned} N_{\text{train}} &= (1 - \texttt{TEST_SIZE})\, N = 0.8N, \\ N_{\text{unlab}} &= \texttt{TEST_SIZE} \cdot \frac{\texttt{TRAIN_SIZE}}{1 - \texttt{TEST_SIZE}}\, N = 0.2 \cdot \frac{0.4}{0.8} N = 0.1N, \\ N_{\text{test}} &= \texttt{TEST_SIZE} \cdot \left(1 - \frac{\texttt{TRAIN_SIZE}}{1 - \texttt{TEST_SIZE}}\right) N = 0.2 \cdot \left(1 - \frac{0.4}{0.8}\right) N = 0.1N. \end{aligned}$$

Concretely, for `adult` and `lawschool` where $N = 2000$, we obtain 1600 labeled training points, 200 unlabeled points, and 200 test points per run. The labeled training split is used to fit the baseline regressor η (linear regression) and the sensitive-attribute classifier (logistic regression), while the unlabeled split is used to fit FairReg and to estimate the OT-based correction; all methods are finally evaluated on the held-out test split.

They use only 2000 points don't know why but i am trying to find out, maybe it's linked to some hyperparameter tuning

Hyperparameters. For FairReg, we use the default hyperparameters from the original code:

- the number of points in the grid is $\sqrt{N_{\text{unlab}}}$ (i.e., 14 for $N_{\text{unlab}} = 200$),
- the entropic regularization parameter is $\beta = \frac{1}{2}\sqrt{N_{\text{unlab}}}\,\log_2 N_{\text{unlab}}$ (i.e., ≈ 54.1 for $N_{\text{unlab}} = 200$), la temperature a un enorme impact sur les perf du modele du coup j'ai pas laissé auto sinon c'etait trop nul
- the DP slack parameter is $\varepsilon = 0.01$.

1 Adult dataset

Adult (Census Income). The `Adult` dataset contains individual-level records from the U.S. census, described by demographic and employment-related attributes such as age, education, marital status, occupation, hours worked per week, and capital gain/loss. The original prediction task is to infer whether an individual earns more than \$50K per year (binary label `income`); in the pipeline of Fairreg, this target is preprocessed into a numeric response. The dataset also

includes sensitive attributes such as `sex` and `race`. y'a aussi des fields de variables categoriques qui ont plein de données manquantes sous forme de "?" qu'on drop au preprocessing

COMMENTAIRE: c'est un peu bizarre ce qu'ils font dans la pipeline parce que pendant le préprocessing au lieu d'utiliser la target de d'habitude ils utilisent l'age c'est pour ça qu ces exemple est pas hyper relevant (of course l'age est pas biaisé en fonction du genre mais au moins ça test sur des vraies valeurs bien confondues les méthodes...)

1.1 Analysis of the dataset

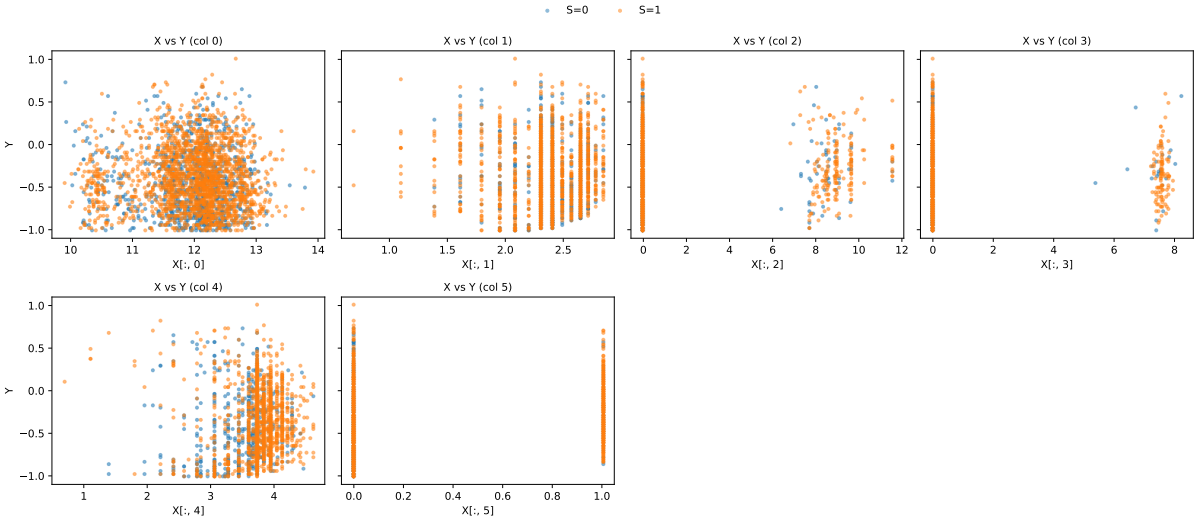


Figure 17: Scatter plot of (X, Y) colored by S (Adult dataset; run 10).

1.2 OT

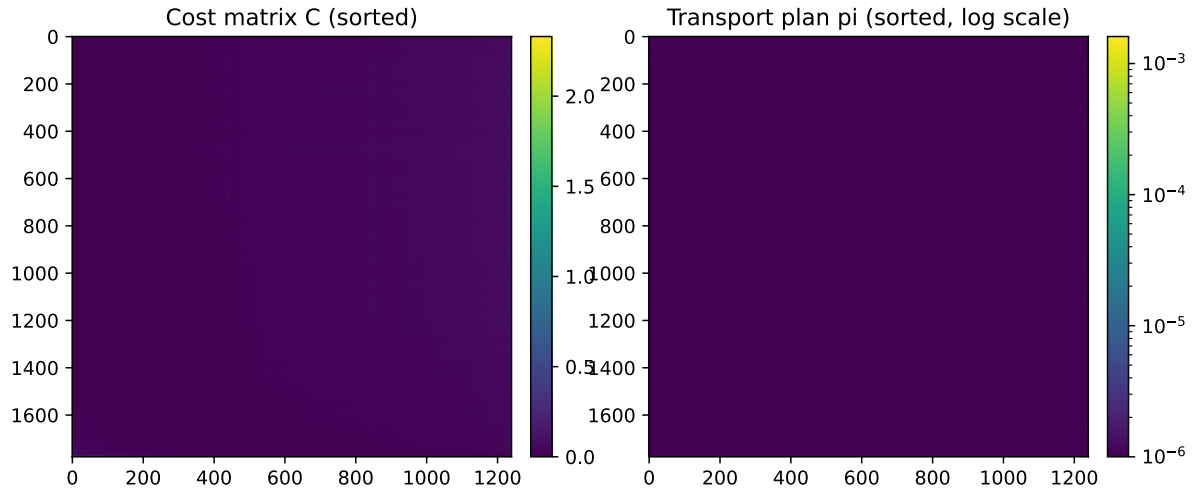


Figure 18: OT plan (Adult dataset; run 10).

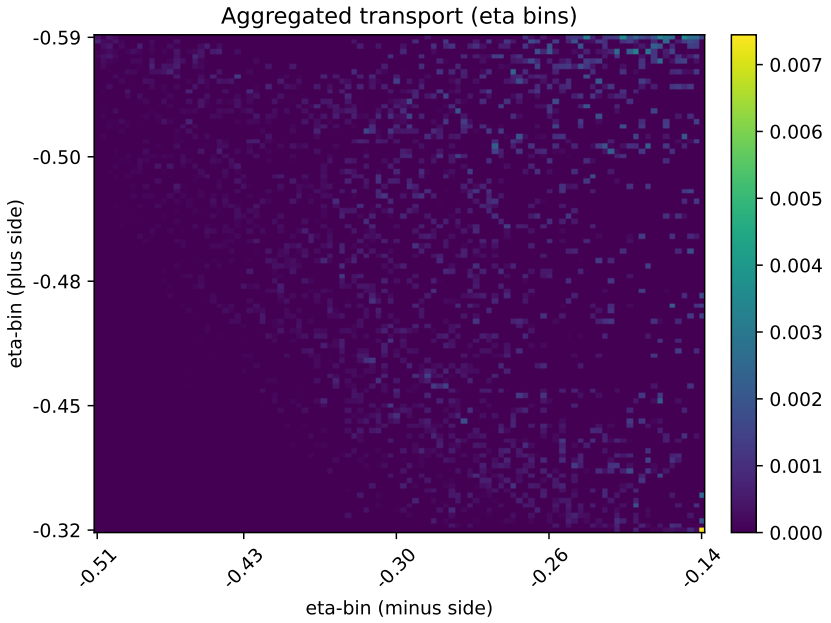


Figure 19: Aggregated OT plan (Adult dataset; run 10).

Quantity	Mean ± Std
`mse_base`	0.0161 ± 0.0005
`mse_fairreg`	0.0162 ± 0.0005
`mse_ot`	0.0170 ± 0.0004
`dp_base_max`	0.1588 ± 0.0101
`dp_fairreg_max`	0.1498 ± 0.0100
`dp_ot_max`	0.0292 ± 0.0096

Table 3: Summary (mean ± std over runs) Adult.

1.3 Hyperparameters tuning

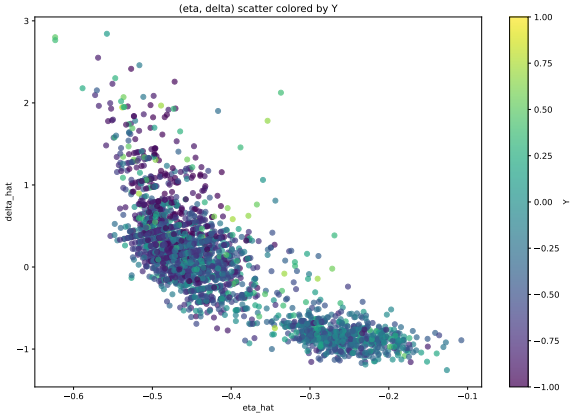
1.4 Unfairness and MSE

1.5 Ordering stability and leakage AUC

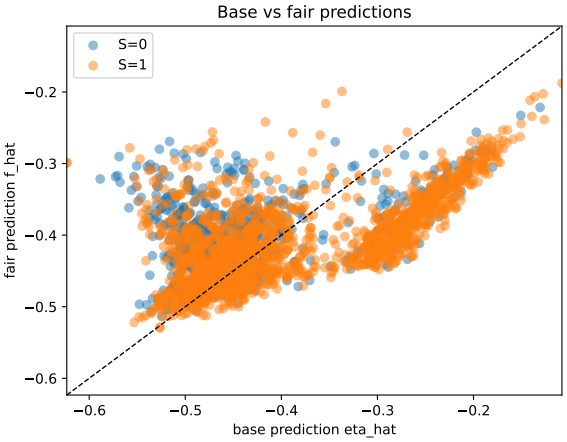
Conclusion: OT better at reducing leakage and preserving order within groups. But the datase is already almost free from leakage so not really relevant.

2 Communities and Crime dataset

Communities and Crime. The `Communities and Crime` dataset is community-level: each example corresponds to a U.S. community characterized by a high-dimensional set of socio-economic, demographic, housing, and policing variables (e.g., racial composition percentages, income and poverty indicators, employment structure, household composition, immigration, housing characteristics, and law-enforcement resources). The canonical target variable is the violent crime rate per population, `ViolentCrimesPerPop`. In many standard versions of the dataset, features are already normalized to the unit interval, and missing entries can occur (depending on the preprocessing). The sensitive attribute is obtained by taking the highest percentage of a racial group in the community (among `White`, `Black`, `Asian`, `Hispanic`, `NativeAmerican`, `Other`) and binarizing it into majority-`White` vs. non-majority-`White` communities.

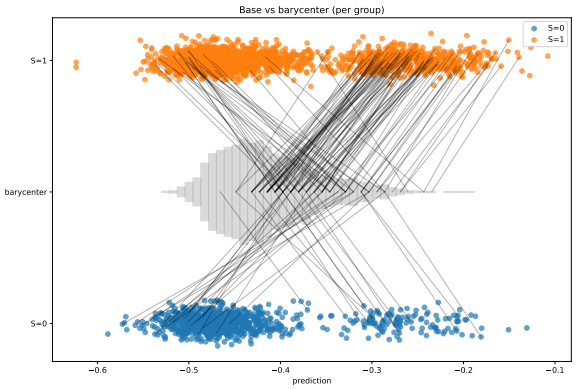


Scatter plot of (X, Y) colored by Δ (Adult dataset; run 10).

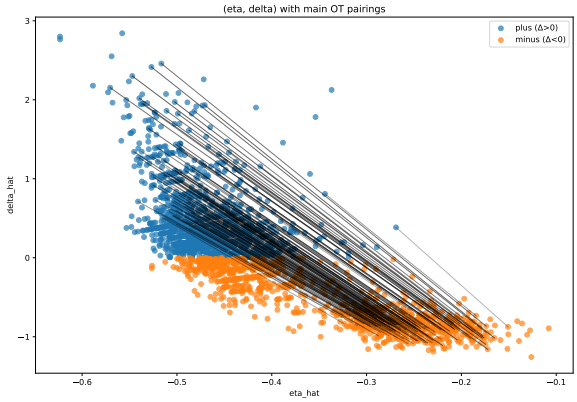


Base vs fair scatter (Adult dataset; run 10).

Figure 20: Side-by-side: colored-by-Δ vs base-vs-fair scatter (Adult dataset; run 10).



Barycenter links. x



Transport lines in (η, Δ) space.

Figure 21: Barycenter links and transport lines (Adult dataset; run 10).

2.1 Analysis of the dataset

Too many columns in X for his dataset so no scatter plot of (X,Y) colored by S. (uncomment to see attempt)

2.2 OT

2.3 Unfairness and MSE

2.4 Ordering stability and leakage AUC

Conclusion: OT better at reducing leakage and preserving order within groups. Here the dataset is very biased so relevant. THe reordering is quite strong on S=0 group (much more than on S=1). Maybe it's because it is actually a merge of multiple communities with different characteristics?

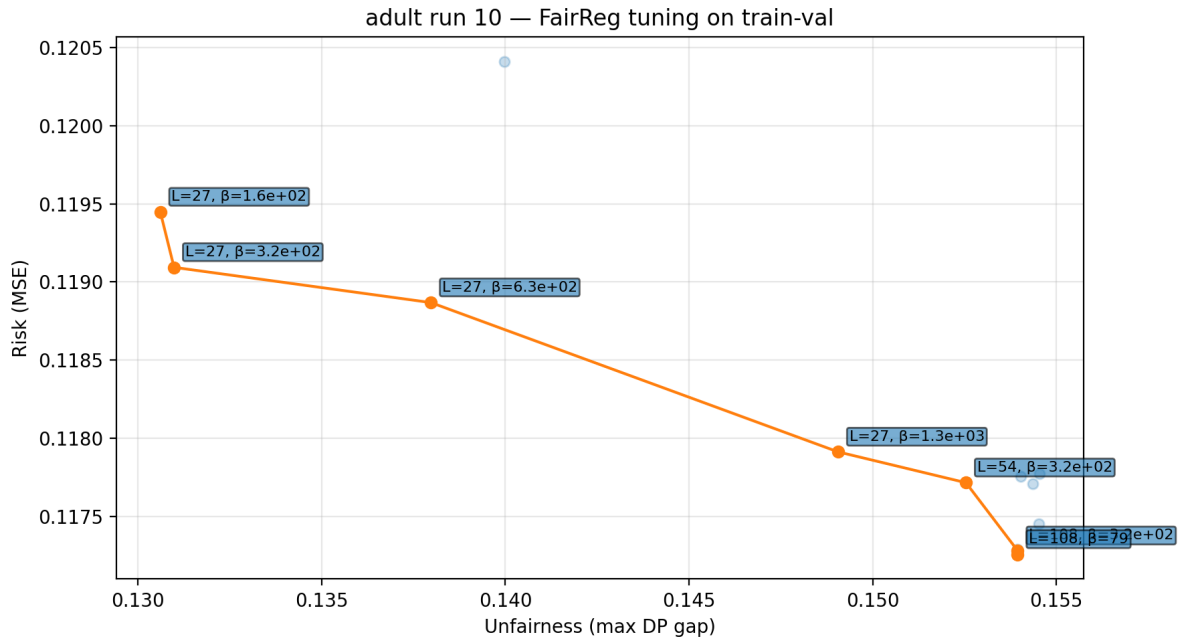


Figure 22: OT tuning curve Adult (run 10).

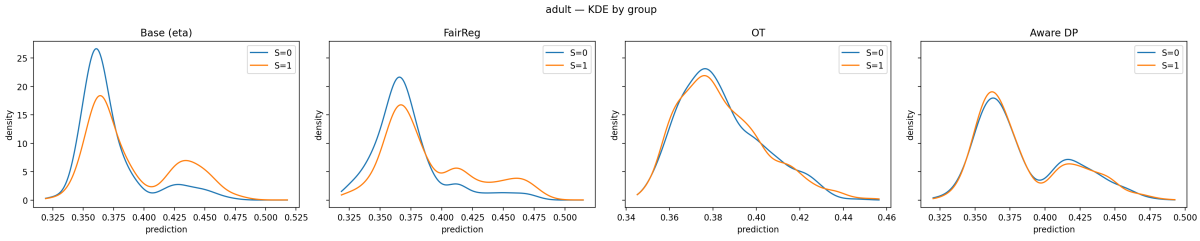


Figure 23: Prediction distributions by group (Adult dataset; run 10).

2.5 Hyperparamters tuning

3 Lawschool dataset

Law School. The `Law School` dataset contains student-level records related to law school admissions and academic outcomes. Covariates include academic indicators such as `lsat` and undergraduate GPA (`ugpa`), along with demographic and background variables such as `race`, `gender`, age, and family income. Outcome variables include standardized grade-point averages (e.g., first-year GPA `zfygpa` and overall GPA `zgpa`) and, in some versions, bar exam passage information (`bar1`). The sensitive attribute is the race "1" (white) vs. "0" (non-white).

3.1 Analysis of the dataset

3.2 OT

3.3 Hyperparamters tuning

3.4 Unfairness and MSE

3.5 Ordering stability and leakage AUC

Subset	Spearman ρ	Kendall τ_b	Mean $\lvert\Delta\text{rank}\rvert$	Top-5% Jaccard
overall	0.447	0.347	0.220	0.408
S=0	0.207	0.163	0.284	0.280
S=1	0.587	0.459	0.181	0.474

Table 4: Ordering stability between $\hat{\eta}$ and corrected scores Adult.

Quantity	AUC
$\text{AUC}(S \mid Y)$	0.567
$\text{AUC}(S \mid \hat{\eta})$	0.653
$\text{AUC}(S \mid \tilde{y})$	0.480

Table 5: Leakage AUC Adult: ability to infer S from a scalar score (values near 0.5 indicate low leakage).

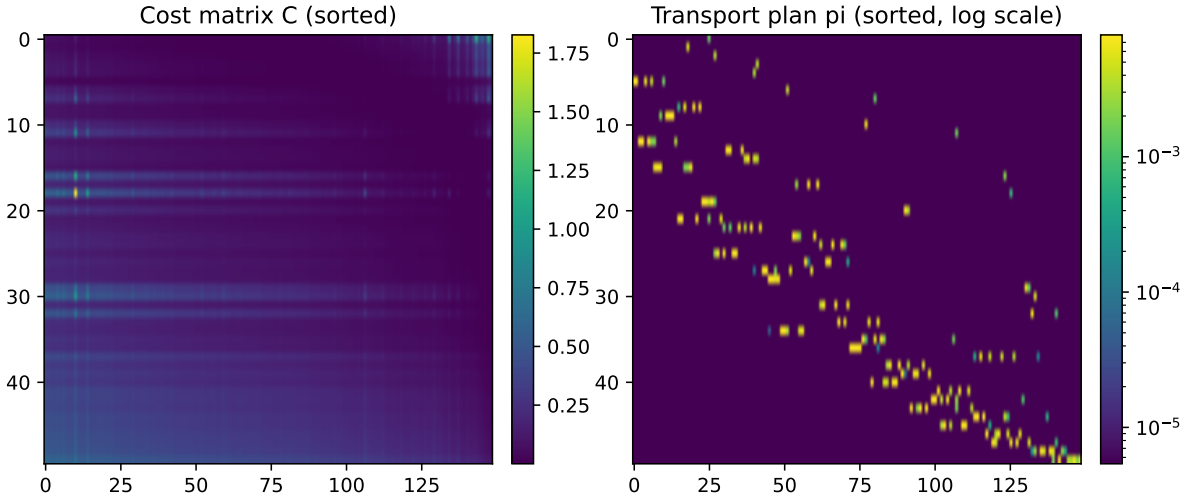


Figure 24: OT plan (Crime dataset; run 10).

Quantity	Mean ± Std
`mse_base`	0.0196 ± 0.0032
`mse_fairreg`	0.0259 ± 0.0037
`mse_ot`	0.0408 ± 0.0059
`dp_base_max`	0.4326 ± 0.0395
`dp_fairreg_max`	0.2616 ± 0.0370
`dp_ot_max`	0.0981 ± 0.0369

Table 6: Summary (mean ± std over runs) Communities and crimes.

Subset	Spearman ρ	Kendall τ_b	Mean $\lvert\Delta\text{rank}\rvert$	Top-5% Jaccard
overall	0.689	0.590	0.167	0.125
S=0	0.526	0.436	0.235	0.333
S=1	0.908	0.813	0.076	0.273

Table 7: Ordering stability between $\hat{\eta}$ and corrected scores.

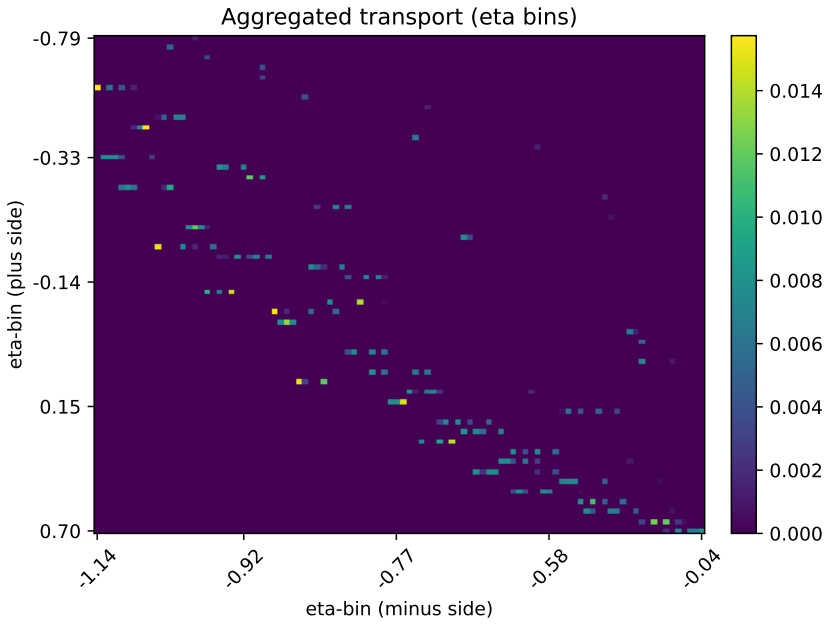
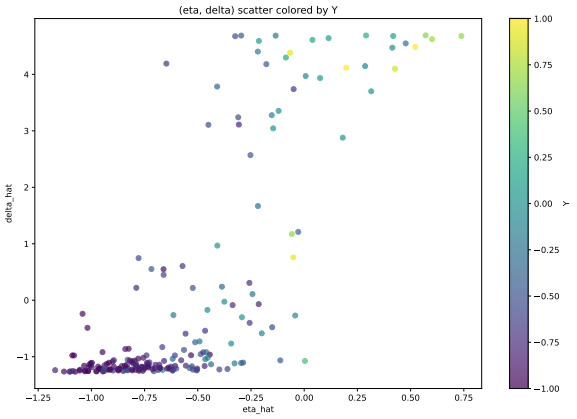
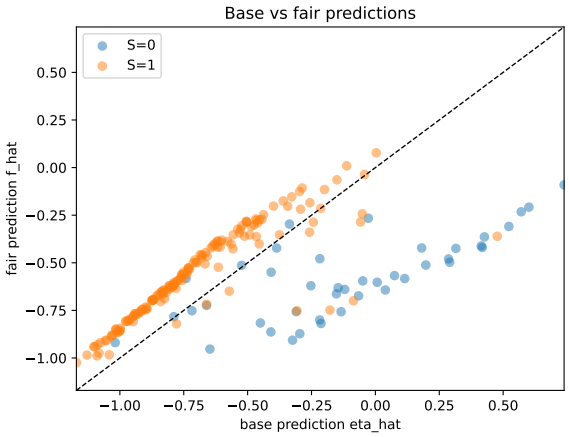


Figure 25: Aggregated OT plan (Crime dataset; run 10).

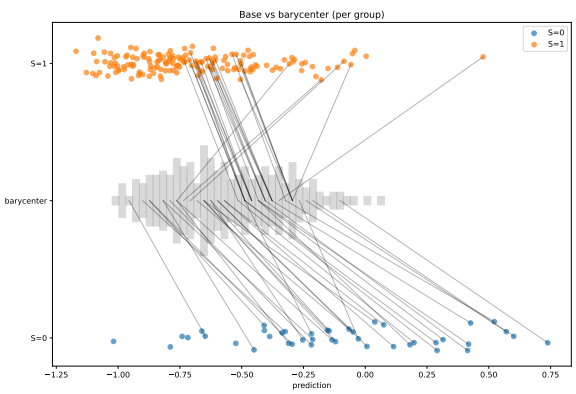


Scatter plot of (X, Y) colored by Δ (Crime dataset; run 10).

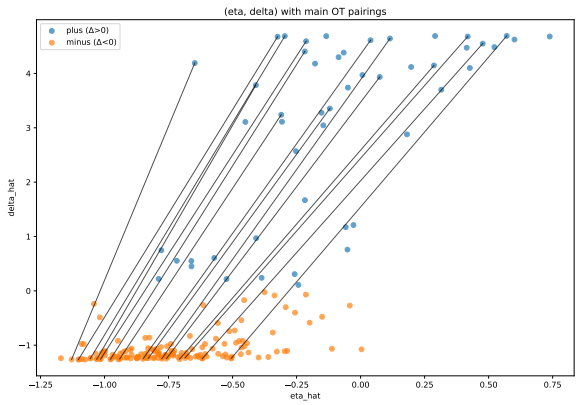


Base vs fair scatter (Crime dataset; run 10).

Figure 26: Side-by-side: colored-by-Δ vs base-vs-fair scatter (Crime dataset; run 10).



Barycenter links.



Transport lines in (η, Δ) space.

Figure 27: Barycenter links and transport lines (Crime dataset; run 10).

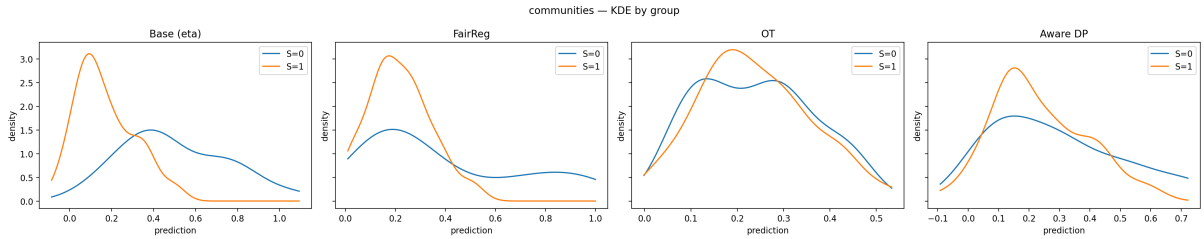


Figure 28: Prediction distributions by group (Crime dataset; run 10).

Quantity	AUC
$\mathrm{AUC}(S \mid Y)$	0.168
$\mathrm{AUC}(S \mid \hat{\eta})$	0.096
$\mathrm{AUC}(S \mid \tilde{y})$	0.477

Table 8: Leakage AUC Comunities and Crime: ability to infer S from a scalar score (values near 0.5 indicate low leakage).

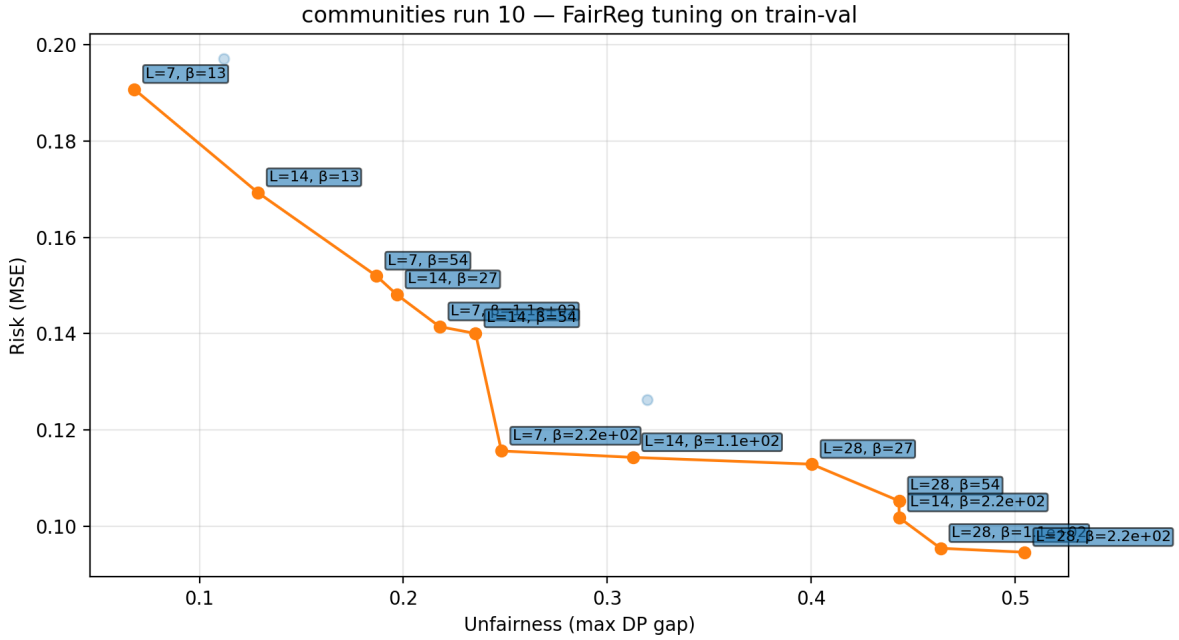


Figure 29: Fairreg tuning curve Communities and Crime (run 10).

Quantity	Mean ± Std
`mse_base`	0.0093 ± 0.0004
`mse_fairreg`	0.0095 ± 0.0004
`mse_ot`	0.0097 ± 0.0004
`dp_base_max`	0.2274 ± 0.0244
`dp_fairreg_max`	0.0969 ± 0.0202
`dp_ot_max`	0.0465 ± 0.0209

Table 9: Summary (mean ± std over runs) Lawschool.

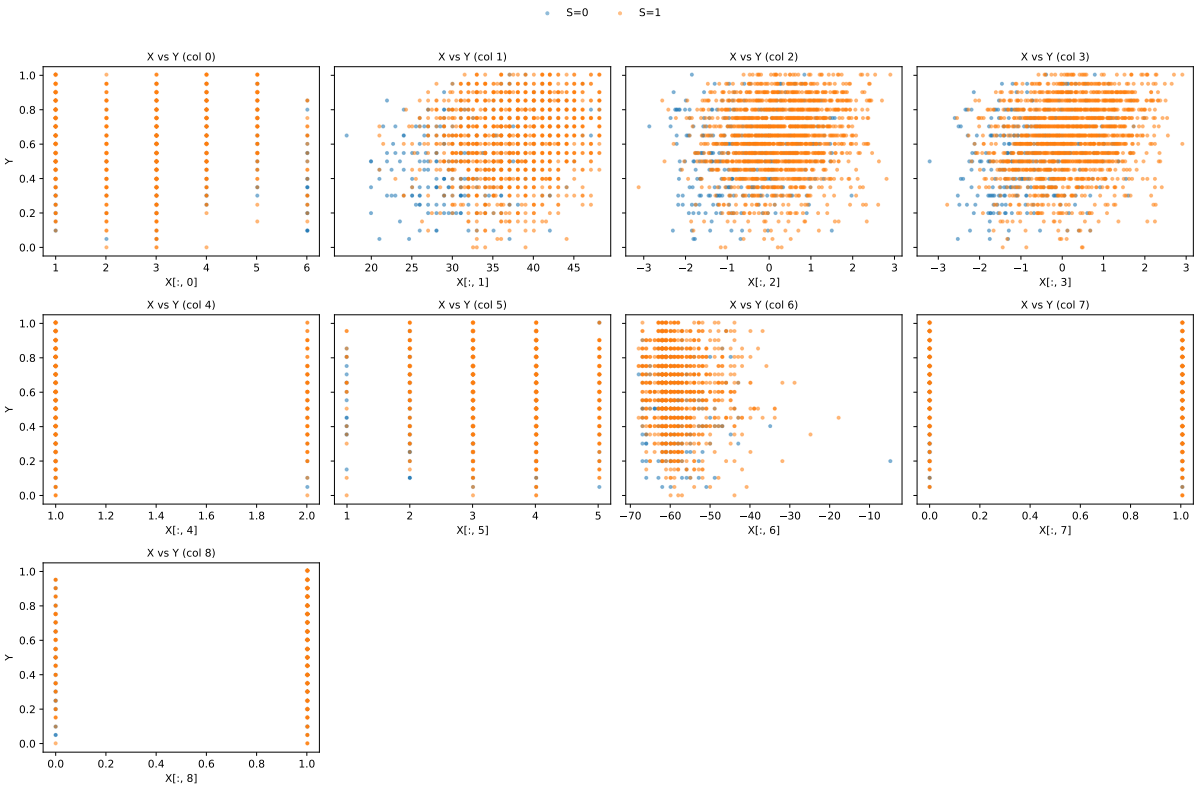


Figure 30: Scatter plot of (X, Y) colored by S (Lawschool dataset; run 10).

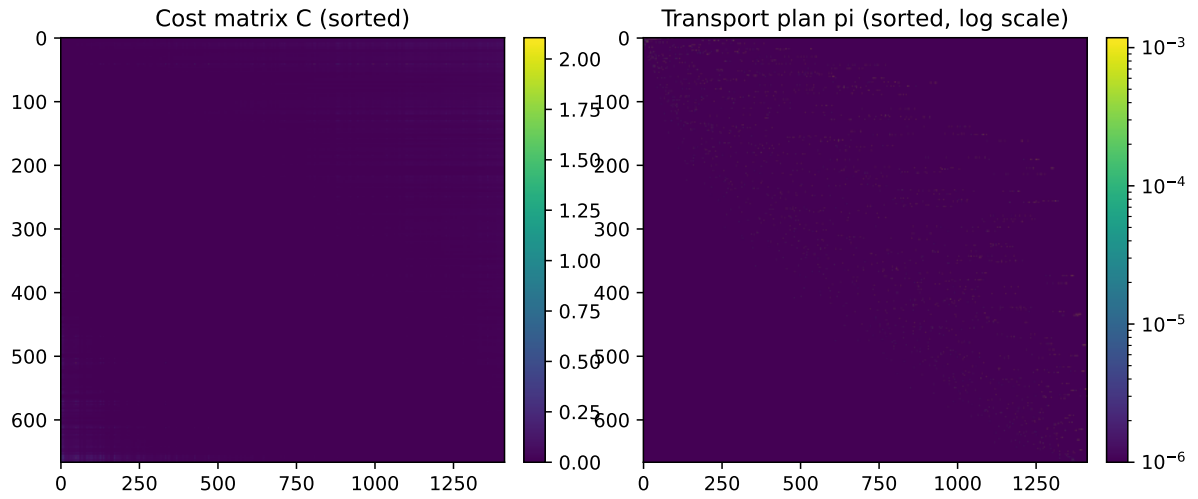


Figure 31: OT plan (Lawschool dataset; run 10).

Subset	Spearman ρ	Kendall τ_b	Mean $\lvert\Delta\text{rank}\rvert$	Top-5% Jaccard
overall	0.900	0.763	0.091	0.667
S=0	0.865	0.677	0.121	1.000
S=1	0.928	0.805	0.076	0.778

Table 10: Ordering stability between $\hat{\eta}$ and corrected scores Lawschool.

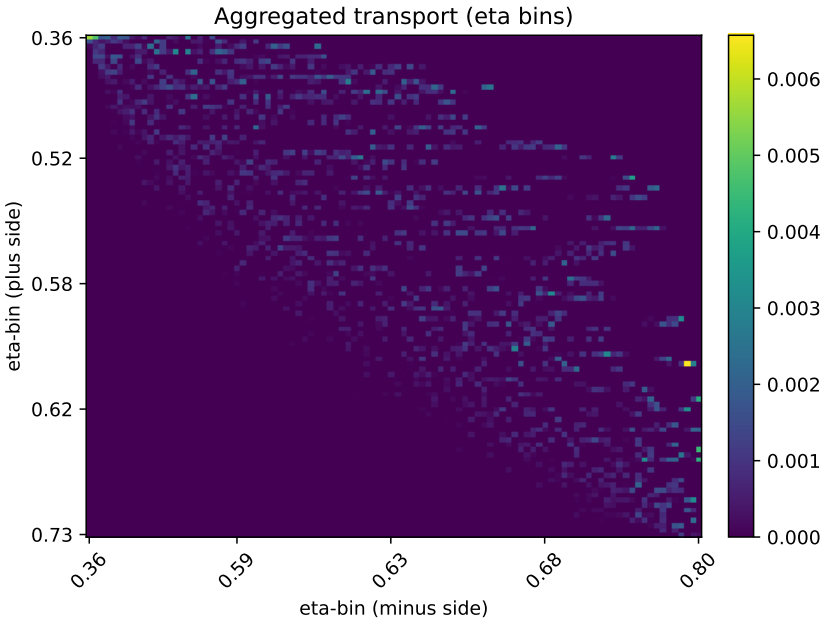
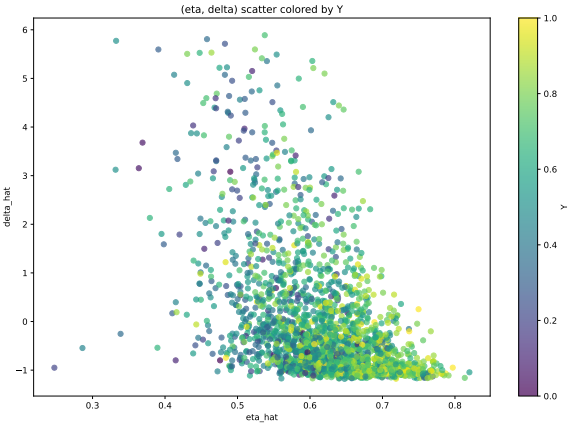
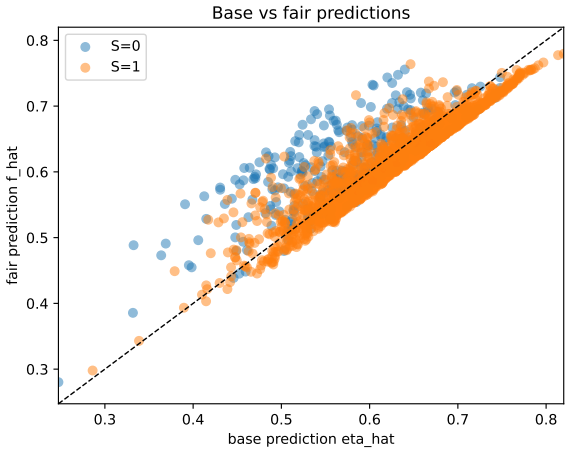


Figure 32: Aggregated OT plan (Lawschool dataset; run 10).

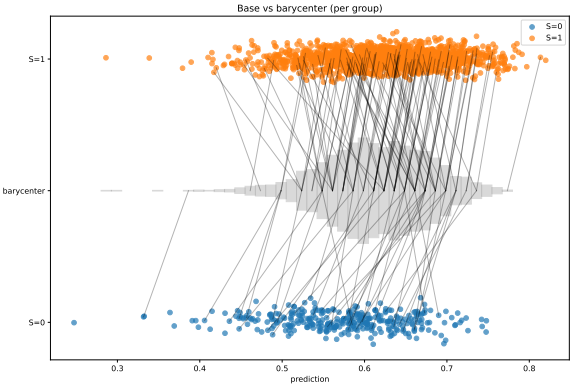


Scatter plot of (X, Y) colored by Δ (Lawschool dataset; run 10).

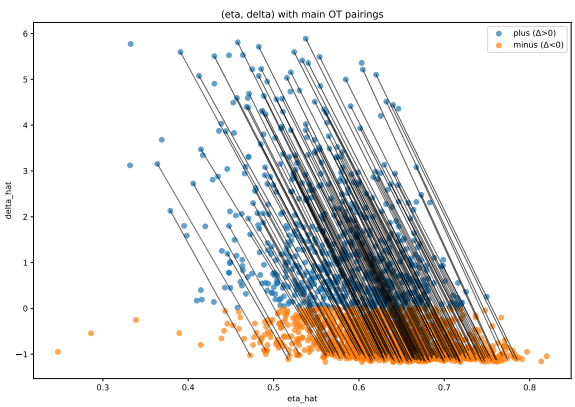


Base vs fair scatter (Lawschool dataset; run 10).

Figure 33: Side-by-side: colored-by-Δ vs base-vs-f air scatter (Lawschool dataset; run 10).



Barycenter links.



Transport lines in (η, Δ) space.

Figure 34: Barycenter links and transport lines (Lawschool dataset; run 10).

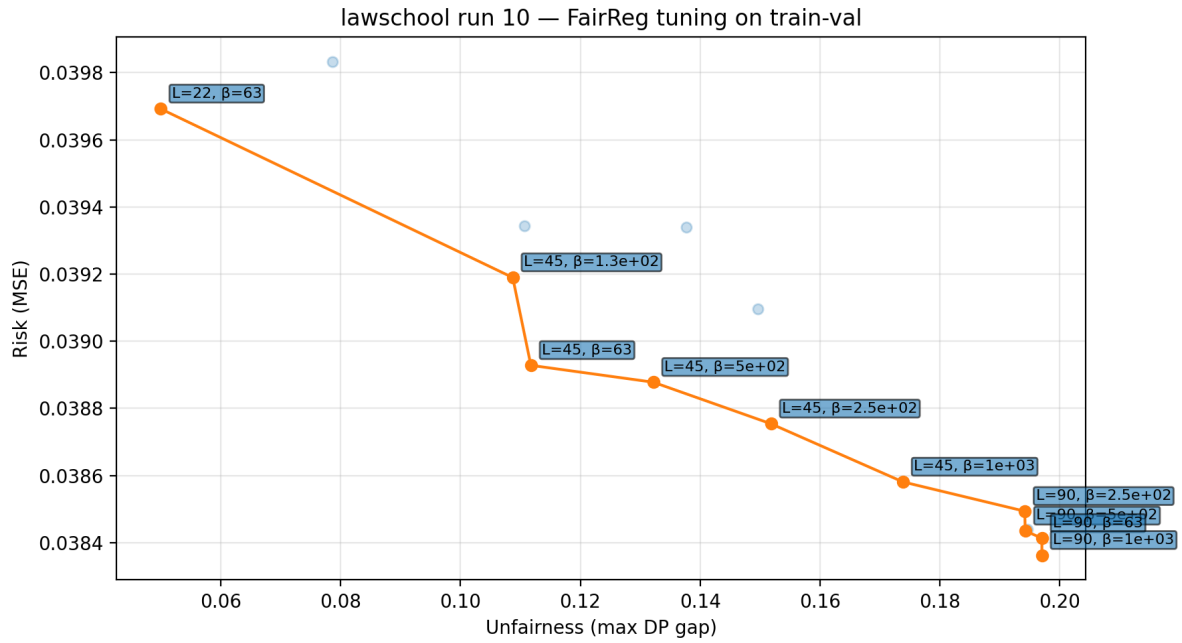


Figure 35: Fairreg tuning curve Lawschool (run 10).

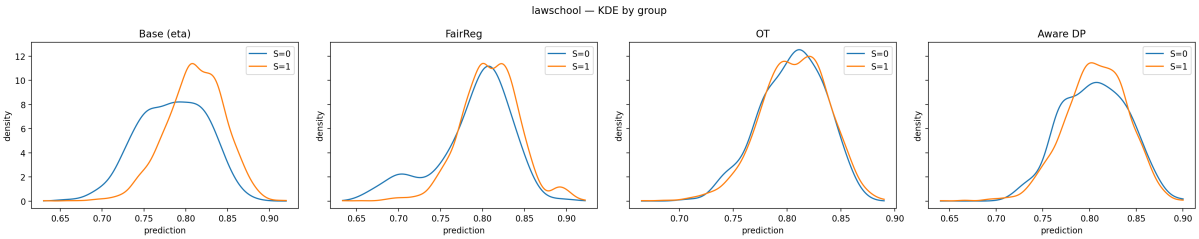


Figure 36: Prediction distributions by group (Lawschool dataset; run 10).

Quantity	AUC
AUC($S \mid Y$)	0.612
AUC($S \mid \hat{\eta}$)	0.637
AUC($S \mid \tilde{y}$)	0.513

Table 11: Leakage AUC Lawschool: ability to infer S from a scalar score (values near 0.5 indicate low leakage).

Part IV
Relaxed unawareness

1 Setting

- Let $\mathcal{X} := \mathbb{R} \times (0, \infty)$ and write $x = (h, a) \in \mathcal{X}$ (with a standing for $|d|$ in the paper).
- Let $\mu^+, \mu^- \in \mathcal{P}(\mathcal{X})$ be given.
- quadratic cost (same as in the paper)

$$c(x, y) \ := \ \frac{(h - y)^2}{a}, \qquad x = (h, a) \in \mathcal{X}, \ y \in \mathbb{R}.$$

- For $\alpha \in \mathcal{P}(\mathcal{X})$ and $\nu \in \mathcal{P}(\mathbb{R})$ set

$$OT_c(\alpha, \nu) \ := \ \inf_{\pi \in \Pi(\alpha, \nu)} \int_{\mathcal{X} \times \mathbb{R}} c(x, y) \, d\pi(x, y),$$

  where $\Pi(\alpha, \nu)$ denotes the set of couplings between α and ν.
- W_2 is the usual 2-Wasserstein distance on $\mathcal{P}(\mathbb{R})$.

2 Relaxed unawareness OT problem

Fix $\lambda > 0$. We consider the relaxed problem

$$\inf_{\nu_1, \nu_2 \in \mathcal{P}(\mathbb{R})} \left\{ OT_c(\mu^+, \nu_1) + OT_c(\mu^-, \nu_2) + \lambda \, W_2^2(\nu_1, \nu_2) \right\}. \tag{P_λ}$$

Proposition 2.1 (Closed-form relax porblem)**.** *Let $x_1 = (h_1, a_1), x_2 = (h_2, a_2) \in \mathcal{X}$. Define the pointwise relaxed cost*

$$C_\lambda(x_1, x_2) \ := \ \inf_{y_1, y_2 \in \mathbb{R}} \left\{ \frac{(h_1 - y_1)^2}{a_1} + \frac{(h_2 - y_2)^2}{a_2} + \lambda (y_1 - y_2)^2 \right\}. \tag{2.1}$$

Then

$$\inf(P_\lambda) \ = \ \inf_{\pi \in \Pi(\mu^+, \mu^-)} \int_{\mathcal{X} \times \mathcal{X}} C_\lambda(x_1, x_2) \, d\pi(x_1, x_2). \tag{2.2}$$

Moreover, the minimizer in (2.1) *is unique and given by*

$$y_1^*(x_1, x_2) = \frac{(1 + \lambda a_2) \, h_1 + \lambda a_1 \, h_2}{1 + \lambda(a_1 + a_2)}, \qquad y_2^*(x_1, x_2) = \frac{\lambda a_2 \, h_1 + (1 + \lambda a_1) \, h_2}{1 + \lambda(a_1 + a_2)}, \tag{2.3}$$

and the optimal value has the closed form

$$C_\lambda(x_1, x_2) \ = \ \frac{\lambda}{1 + \lambda(a_1 + a_2)} \, (h_1 - h_2)^2. \tag{2.4}$$

Proof. **Step 1: comupling and gluing lemma.** We expand each term in (P_λ) with couplings:

$$OT_c(\mu^+, \nu_1) = \inf_{\pi_1 \in \Pi(\mu^+, \nu_1)} \int c(x_1, y_1) \, d\pi_1, \quad OT_c(\mu^-, \nu_2) = \inf_{\pi_2 \in \Pi(\mu^-, \nu_2)} \int c(x_2, y_2) \, d\pi_2,$$

$$W_2^2(\nu_1, \nu_2) = \inf_{\gamma \in \Pi(\nu_1, \nu_2)} \int (y_1 - y_2)^2 \, d\gamma.$$

Hence

$$\inf(P_\lambda) = \inf_{\nu_1, \nu_2} \ \inf_{\substack{\pi_1 \in \Pi(\mu^+, \nu_1) \\ \pi_2 \in \Pi(\mu^-, \nu_2) \\ \gamma \in \Pi(\nu_1, \nu_2)}} \left\{ \int c(x_1, y_1) \, d\pi_1 + \int c(x_2, y_2) \, d\pi_2 + \lambda \int (y_1 - y_2)^2 \, d\gamma \right\}.$$

Gluing lemma. Let $\pi_1 \in \Pi(\mu^+, \nu_1)$, $\gamma \in \Pi(\nu_1, \nu_2)$, and $\pi_2 \in \Pi(\mu^-, \nu_2)$, and assume the *shared marginals are consistent*, i.e.

$$(\mathrm{pr}_{y_1})_\# \pi_1 = (\mathrm{pr}_{y_1})_\# \gamma = \nu_1, \qquad (\mathrm{pr}_{y_2})_\# \gamma = (\mathrm{pr}_{y_2})_\# \pi_2 = \nu_2.$$

One may view π_1, γ, π_2 as the laws of random pairs (X_1, Y_1), (Y_1, Y_2) and (X_2, Y_2) whose overlaps Y_1 and Y_2 have the same distributions. By the gluing lemma (see, e.g., Lemma 5.5 in Santambrogio (2015)), there exists a probability measure Π on $\mathcal{X} \times \mathcal{X} \times \mathbb{R} \times \mathbb{R}$ such that

$$(\mathrm{pr}_{x_1,y_1})_\# \Pi = \pi_1, \qquad (\mathrm{pr}_{y_1,y_2})_\# \Pi = \gamma, \qquad (\mathrm{pr}_{x_2,y_2})_\# \Pi = \pi_2.$$

Conversely, any such Π determines feasible couplings by marginalization. Hence the relaxed objective can be expressed using a single joint law:

$$\inf(P_\lambda) = \inf_{\Pi \in \mathcal{A}} \int_{\mathcal{X}^2 \times \mathbb{R}^2} \Big(c(x_1, y_1) + c(x_2, y_2) + \lambda (y_1 - y_2)^2 \Big)\, d\Pi, \tag{2.5}$$

where $\mathcal{A}$ is the set of probability measures Π on $\mathcal{X}^2 \times \mathbb{R}^2$ such that $(\mathrm{pr}_{x_1})_\# \Pi = \mu^+$ and $(\mathrm{pr}_{x_2})_\# \Pi = \mu^-$.

Step 2: reduce to pointwize optimization. Let $\Pi \in \mathcal{A}$ and let $\pi := (\mathrm{pr}_{x_1,x_2})_\# \Pi \in \Pi(\mu^+, \mu^-)$ (the (X_1, X_2) marginal). We can write

$$\pi_1(dx_1, dy_1) = \nu_1(dy_1) = \nu_1(dy_1) K_1(dx_1|y_1),$$
$$\pi_2(dx_2, dy_2) = \nu_2(dy_2) = \nu_2(dy_2) K_2(dx_2|y_2),$$

where $K_i(\cdot|y_i)$ are the corresponding conditional distributions of X_i knowing $Y_i = y_i$. Then

$$\Pi(dx_1, dx_2, dy_1, dy_2) = \gamma(dy_1, dy_2)\, K_1(dx_1|y_1)\, K_2(dx_2|y_2).$$

on tire $(Y_1, Y_2) \sim \gamma$, puis $X_1|Y_1 \sim K_1(\cdot|Y_1)$ et $X_2|Y_2 \sim K_2(\cdot|Y_2)$ à partir de π_i indépendamment suivant y_1, Y_2.
Thereforen, there exists a Markov kernel (i.e., measurable map) K_{x_1,x_2} on $\mathbb{R}^2$ such that

$$\Pi(dx_1, dx_2, dy_1, dy_2) = \pi(dx_1, dx_2)\, K_{x_1,x_2}(dy_1, dy_2).$$

Write the integrand as

$$L(x_1, x_2; y_1, y_2) := c(x_1, y_1) + c(x_2, y_2) + \lambda (y_1 - y_2)^2.$$

Then (2.5) becomes

$$\int_{\mathcal{X}^2} \left(\int_{\mathbb{R}^2} L(x_1, x_2; y_1, y_2)\, dK_{x_1,x_2}(y_1, y_2) \right) d\pi(x_1, x_2).$$

For each fixed (x_1, x_2), the map $(y_1, y_2) \mapsto L(x_1, x_2; y_1, y_2)$ is a strictly convex quadratic. Hence, for any probability measure K on $\mathbb{R}^2$,

$$\int L\, dK \;\geq\; \inf_{(y_1,y_2) \in \mathbb{R}^2} L(x_1, x_2; y_1, y_2),$$

with equality achieved by choosing K as a Dirac mass at the unique minimizer $\delta_{(y_1^*, y_2*)}$. Then for a fixed couple (x_1, x_2) the conditional distribution of (Y_1, Y_2) given $(X_1, X_2) = (x_1, x_2)$ that minimizes the inner integral is the Dirac mass at the unique minimizer of $L(x_1, x_2; \cdot, \cdot)$. We have

$$C_\lambda(x_1, x_2) := \inf_{y_1, y_2 \in \mathbb{R}} L(x_1, x_2; y_1, y_2)$$

Therefore, for a fixed $\pi \in \Pi(\mu^+, \mu^-)$, the optimization over $\Pi \in \mathcal{A}$ with this marginal reduces tp an optimization over the kernels $\{K_{x_1,x_2}\}$, and we have

$$\int_{\mathcal{X}^2} \left(\underbrace{\int_{\mathbb{R}^2} L(x_1, x_2; \cdot)\, dK_{x_1,x_2}(y_1, y_2)}_{\text{depends only on } K_{x_1,x_2}} \right) d\pi(x_1, x_2).$$

Thus we can minimize the inner integral pointwise. At a fixed $\pi \in \Pi(\mu^+, \mu^-)$ we have that the best kernel choise is

$$K_{x_1,x_2} = \delta_{(y_1^*(x_1,x_2)\,,\, y_2^*(x_1,x_2))},$$

where $(y_1^*(x_1, x_2), y_2^*(x_1, x_2))$ is the unique minimizer of $L(x_1, x_2; \cdot, \cdot)$. We went from an optimization over $\Pi \in \mathcal{A}$ with fixed $(\mathrm{pr}_{x_1,x_2})_\# \Pi = \pi$ to a pointwise minimization for each π where we know the optimal kernel. Hence

$$\inf_{\Pi \in \mathcal{A}:\ (\mathrm{pr}_{x_1,x_2})_\# \Pi = \pi} \int L\, d\Pi \int_{\mathcal{X}^2} \underbrace{(L(x_1, x_2; y_1, y_2))}_{C_\lambda(x_1,x_2)}\, d\pi(x_1, x_2).$$

where C_λ is defined in (2.1). Taking the infimum over $\pi \in \Pi(\mu^+, \mu^-)$ yields (2.2).

Step 3: closed form. Fix $x_i = (h_i, a_i)$ and minimize

$$\Phi(y_1, y_2) = \frac{(h_1 - y_1)^2}{a_1} + \frac{(h_2 - y_2)^2}{a_2} + \lambda(y_1 - y_2)^2.$$

First-order optimality gives

$$\begin{cases} \frac{2(y_1 - h_1)}{a_1} + 2\lambda(y_1 - y_2) & = 0, \\ \frac{2(y_2 - h_2)}{a_2} - 2\lambda(y_1 - y_2) & = 0. \end{cases}$$

Solving the system yields (2.3). Let $u := y_1^* - y_2^*$. From the optimality conditions we also have

$$\begin{cases} y_1^* - h_1 & = -\lambda a_1 u, \\ y_2^* - h_2 & = \lambda a_2 u, \end{cases}$$

and subtracting these identities gives

$$u = (y_1^* - y_2^*) = (h_1 - h_2) - \lambda(a_1 + a_2)u,$$

and then

$$u = \frac{h_1 - h_2}{1 + \lambda(a_1 + a_2)}.$$

Plugging into Φ at optimum:

$$\begin{aligned} C_\lambda(x_1, x_2) &= \Phi(y_1^*, y_2^*), \\ &= \frac{(\lambda a_1 u)^2}{a_1} + \frac{(\lambda a_2 u)^2}{a_2} + \lambda u^2, \\ &= \lambda u^2 \big(1 + \lambda(a_1 + a_2)\big), \end{aligned}$$

hence

$$\begin{aligned} C_\lambda(x_1, x_2) &= \lambda \frac{(h_1 - h_2)^2}{(1 + \lambda(a_1 + a_2))^2} \,(1 + \lambda(a_1 + a_2)), \\ &= \frac{\lambda}{1 + \lambda(a_1 + a_2)} (h_1 - h_2)^2, \end{aligned}$$

which is (2.4). □

Remark 2.2 (Sanity check for compatiblity wih he unrelaxed solution). Consider the relaxed objective (P_λ) and its reduced OT form (2.2) with

$$C_\lambda(x_1, x_2) = \inf_{y_1, y_2} \Big\{ c(x_1, y_1) + c(x_2, y_2) + \lambda (y_1 - y_2)^2 \Big\}.$$

Let $x_i = (h_i, a_i)$ with $a_i > 0$. The inner minimization admits the explicit minimizer $(y_1^*(x_1, x_2), y_2^*(x_1, x_2))$ given in (2.3), and one checks that

$$y_1^*(x_1, x_2) - y_2^*(x_1, x_2) = \frac{h_1 - h_2}{1 + \lambda(a_1 + a_2)} \xrightarrow[\lambda \to \infty]{} 0, \tag{2.6}$$

and moreover

$$y_1^*(x_1, x_2),\ y_2^*(x_1, x_2) \xrightarrow[\lambda \to \infty]{} \frac{a_2 h_1 + a_1 h_2}{a_1 + a_2} =: y^\star(x_1, x_2), \tag{2.7}$$

where $y^\star(x_1, x_2)$ is exactly the minimizer of the unrelaxed two-to-one collapse $\inf_y \{c(x_1, y) + c(x_2, y)\}$.

At the level of pairwise costs, the closed form (2.4) yields the pointwise convergence

$$C_\lambda(x_1, x_2) = \frac{\lambda}{1 + \lambda(a_1 + a_2)} (h_1 - h_2)^2 \xrightarrow[\lambda \to \infty]{} \frac{(h_1 - h_2)^2}{a_1 + a_2} =: C_\infty(x_1, x_2),$$

which is precisely the unrelaxed OT cost. In particular, if V_λ denotes the optimal value of (2.2), then V_λ is nondecreasing in λ and satisfies $V_\lambda \uparrow V_\infty$ as $\lambda \to \infty$, where V_∞ is the optimal value of the unrelaxed problem.

Finally, let $\pi_\lambda^* \in \Pi(\mu^+, \mu^-)$ be an optimal coupling for the reduced problem (2.2), and define the associated optimal outputs by

$$Y_{1,\lambda} := T_{1,\lambda}(X_1, X_2), \qquad Y_{2,\lambda} := T_{2,\lambda}(X_1, X_2), \qquad (X_1, X_2) \sim \pi_\lambda^*,$$

so that $\nu_{1,\lambda}^* = \mathcal{L}(Y_{1,\lambda}) = (T_{1,\lambda})_\# \pi_\lambda^*$ and $\nu_{2,\lambda}^* = \mathcal{L}(Y_{2,\lambda}) = (T_{2,\lambda})_\# \pi_\lambda^*$. Consider moreover the coupling between these two output laws induced by the same construction,

$$\gamma_\lambda := (T_{1,\lambda}, T_{2,\lambda})_\# \pi_\lambda^* \in \Pi(\nu_{1,\lambda}^*, \nu_{2,\lambda}^*).$$

By definition of W_2 we have

$$\begin{aligned} W_2^2(\nu_{1,\lambda}^*, \nu_{2,\lambda}^*) \ &\le \ \int_{\mathbb{R}^2} (y_1 - y_2)^2 \, d\gamma_\lambda(y_1, y_2), \\ &= \ \mathbb{E}_{\pi_\lambda^*}\big[(Y_{1,\lambda} - Y_{2,\lambda})^2\big]. \end{aligned}$$

Using the explicit formula (2.6), we obtain that $Y_{1,\lambda} - Y_{2,\lambda} \to 0$ pointwise in (x_1, x_2) as $\lambda \to \infty$, and hence (under the usual integrability ensured here by the quadratic structure) the right-hand side vanishes as $\lambda \to \infty$. In particular, the two output laws become arbitrarily close in W_2:

$$W_2(\nu_{1,\lambda}^*, \nu_{2,\lambda}^*) \xrightarrow[\lambda \to \infty]{} 0.$$

Therefore the relaxed solutions asymptotically enforce a single common output distribution; if the unrelaxed barycenter is unique, this common limit must coincide with the unrelaxed barycenter law.

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